

Supplementary Materials for
**Ubiquitin receptor–mediated, ubiquitin-independent targeted protein
degradation via 26S proteasomes**

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Sci. Adv. **12**, eadh0227 (2026)
DOI: 10.1126/sciadv.eadh0227

The PDF file includes:

Figs. S1 to S7
Table S1
Legend for data S1

Other Supplementary Material for this manuscript includes the following:

Data S1

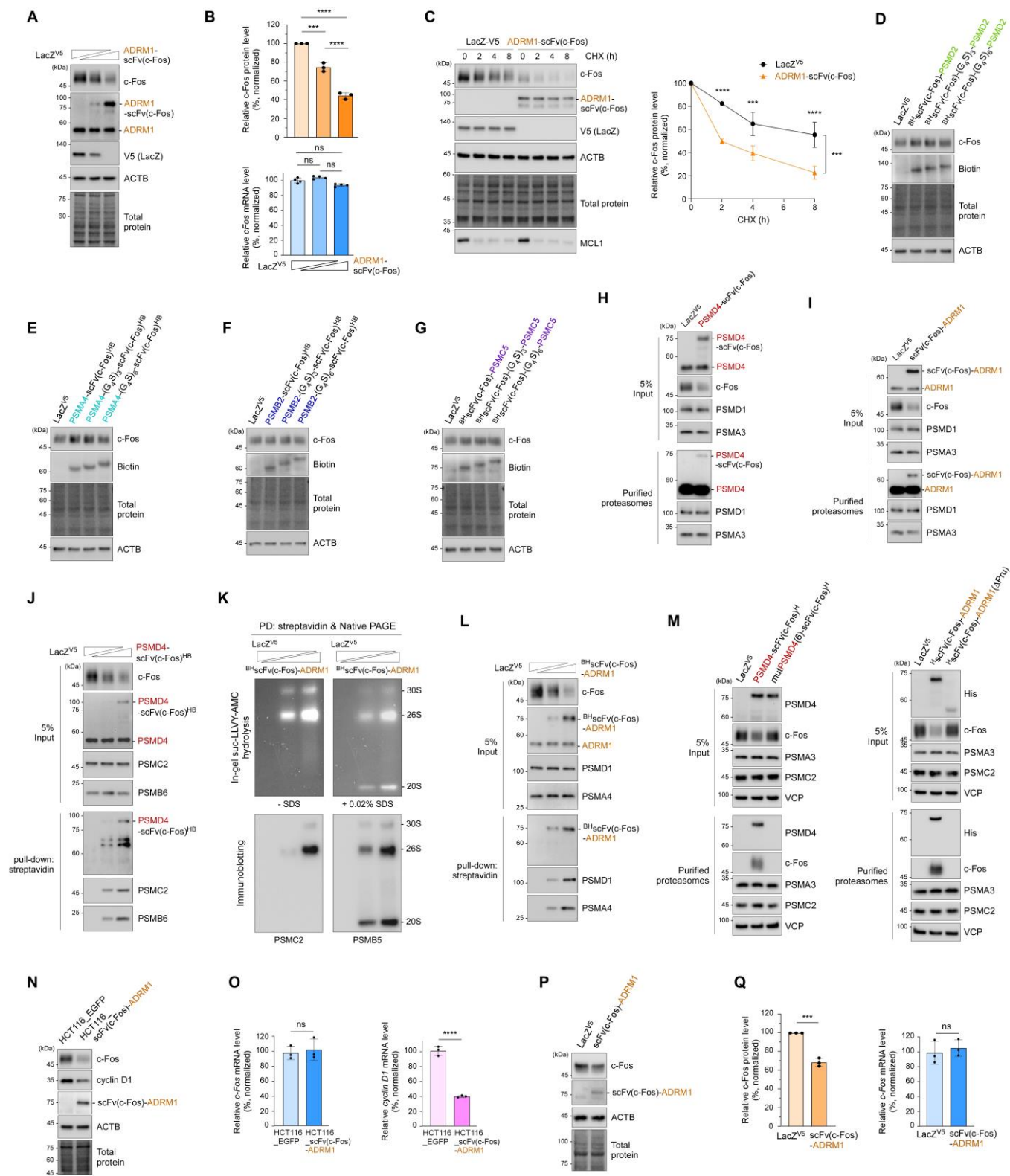


Fig. S1. Protea-Tac degraders promote c-Fos degradation and proteasome remodeling.

Related to Figure 1.

(A) A549 cells were co-transfected with plasmids expressing LacZ^{V5} controls (1, 0.7, and 0 μ g) or ADRM1-scFv(c-Fos) degraders (0, 0.3, and 1 μ g), along with plasmids expressing target protein c-Fos (0.5 μ g). After 64 h, whole-cell lysates (WCLs) were prepared and analyzed by SDS-PAGE/immunoblotting (IB) with indicated antibodies, followed by total protein staining. Representative IB results from three independent experiments are shown. (B) (*top*) Quantification of c-Fos band intensities from (A), normalized to total protein signals. Data represent mean $\pm$ SD ($N = 3$). *** $P < 0.001$, **** $P < 0.0001$ (one-way ANOVA with Tukey's post hoc test). (*bottom*) Total RNA was extracted and analyzed by quantitative RT-PCR (qRT-PCR) using primers targeting *c-Fos* and *GAPDH* (as a normalization control) mRNA. Data represent the mean $\pm$ SD from four independent experiments ($N = 4$). ns, not significant (one-way ANOVA with Tukey's multiple comparison test). (C) Same as (A), except that chase experiments were performed by adding 80 μ g/mL cycloheximide (CHX) and harvesting samples at 2, 4, or 8 h. c-Fos levels were normalized to total protein and to the values at 0 h post-CHX treatment (set as 100%) of each group (LacZ^{V5} controls / ADRM1-scFv(c-Fos) degraders). Average values of normalized c-Fos levels (mean $\pm$ SD) from three independent experiments are shown ($N = 3$, *** $P < 0.001$, **** $P < 0.0001$ from two-way ANOVA followed by Sidak's multiple comparison test). (D–G) PSMD2 (D), PSMA4 (E), PSMB2 (F), and PSMC5 (G) were fused to scFv(c-Fos) using linkers of varying lengths and evaluated for TPD activity. (H) HEK293 cells stably expressing biotin-tagged PSMB2 were transfected with either LacZ^{V5} or PSMD4-scFv(c-Fos) together with c-Fos-expressing plasmids for 64 h. Affinity purification of 26S proteasomes was conducted, and the samples were analyzed by IB. (I) The ADRM1-based degrader (scFv(c-Fos)-ADRM1) was evaluated in the same methods as in (H). (J) PSMD4-scFv(c-Fos) degraders with C-terminal hexahistidine-biotin (HB) tags were transiently expressed for streptavidin affinity purification. Pulldown samples and WCLs were examined by denaturing SDS-PAGE/IB using antibodies against c-Fos and proteasome subunits. (K) Similar to (J), except that N-terminally biotin-hexahistidine (BH)-tagged scFv(c-Fos)-ADRM1 was evaluated. Samples were analyzed by non-denaturing (native) PAGE, followed by in-gel suc-LLVY-AMC hydrolysis visualization (*top*) and IB with antibodies against the CP (PSMB5) and RP (PSMC2) subcomplexes (*bottom*). (L) Similar to (J), except N-terminally biotin-hexahistidine (BH)-tagged scFv(c-Fos)-ADRM1 was analyzed. (M) Protea-Tac degraders harboring mutant PSMD4 (where six key residues in its VWA domain were substituted with alanine; D11A, S13A, T15A, I84A, G115A, and S116A) and Pru-domain-truncated ADRM1 were compared with their cognate Protea-Tac degraders in terms of c-Fos degradation and proteasomal incorporation. (N) Endogenous levels of c-Fos and cyclin D1 were compared in HCT116 cells stably overexpressing EGFP or scFv(c-Fos)-ADRM1. (O) As in (N), except mRNA levels of endogenous *c-Fos* (*top*) and *cyclin D1* (*bottom*) were measured by qRT-PCR and normalized to *GAPDH*. **** $P < 0.0001$ ($N = 3$, two-tailed Student's *t*-test). (P) HCT116 cells were transiently transfected with either LacZ^{V5} or scFv(c-Fos)-ADRM1 for 48 h, followed by IB analysis to determine c-Fos levels. Representative IB results from three independent experiments are shown. (Q) (*left*) Quantification of c-Fos protein levels in (P), normalized to total protein signals. Data represent the mean $\pm$ SD from three independent experiments. *** $P < 0.001$ ($N = 3$, two-tailed Student's *t*-test). (*right*) As in (P), but c-Fos mRNA levels were quantified by qRT-PCR and normalized to *GAPDH*. Data are presented as the mean $\pm$ SD. ns, not significant ($N = 3$, two-tailed Student's *t*-test).

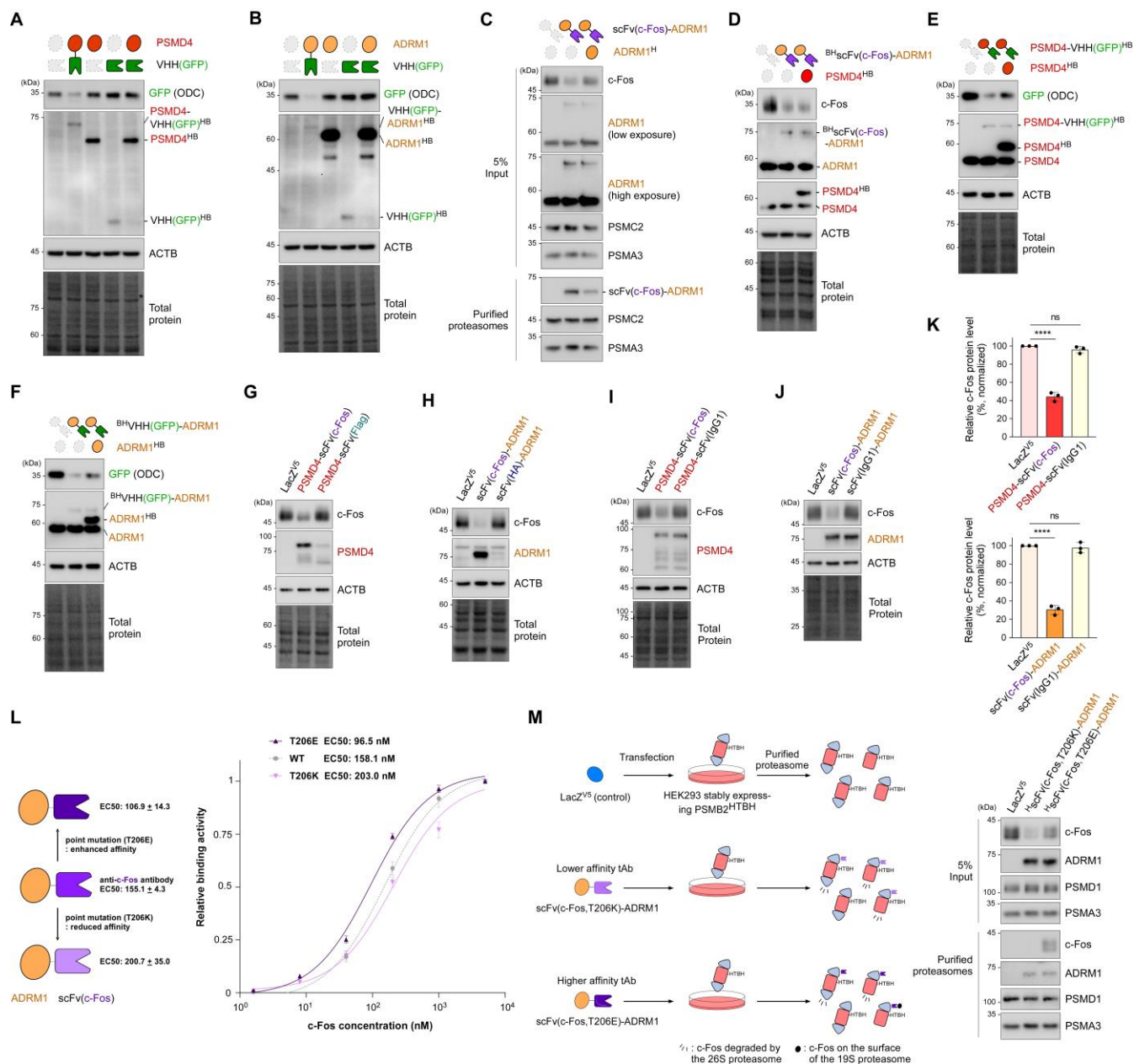


Fig. S2. Protea-Tac is a modular system requiring fused UbR and tAb components. Related to Figure 2.

(A) A549 cells were co-transfected with plasmids expressing PSMD4, VHH(GFP), either individually, or as a PSMD4-VHH (GFP) chimera, along with plasmids expressing GFP-ODC. LacZ^{V5}-expressing plasmids were used as a control to maintain a total of 1.5 µg plasmid DNA per condition. After 64 h, WCLs were prepared and subjected to SDS-PAGE/IB with the indicated antibodies, followed by total protein staining. (B) As in (A), except that ADRM1 was used as the UbR rather than PSMD4. (C) HEK293 cells stably expressing biotin-tagged PSMB2 were transfected with scFv(c-Fos)-ADRM1 and c-Fos-expressing plasmids in the presence or absence

of hexahistidine-tagged ADRM1 (ADRM1^H) to assess competitive incorporation into proteasomes and target degradation efficiency. Proteasomes were affinity-purified, and the samples were analyzed by denaturing SDS-PAGE/IB using antibodies against c-Fos and proteasome subunits. **(D)** As in **(C)**, except that PSMD4 (non-cognate UbR) was co-expressed with scFv(c-Fos)-ADRM1 in A549 cells. **(E and F)** Similar to **(D)**, but PSMD4 or ADRM1 was co-expressed with their corresponding Protea-Tac constructs (PSMD4-VHH(GFP) or VHH(GFP)-ADRM1) to evaluate their competitive effects on GFP-ODC degradation. **(G and H)** Protea-Tac degraders with scFv(Flag) and scFv(HA) were compared with PSMD4-scFv(c-Fos) and scFv(c-Fos)-ADRM1 chimeras, respectively, in terms of c-Fos degradation. **(I and J)** Protea-Tac chimeras of PSMD4 and ADRM1 fused to a mutant scFv construct whose CDR regions were substituted with that of human IgG1 were examined. **(K)** *(top)* Quantification of c-Fos band intensities from **(I)**, normalized to total protein. *(bottom)* c-Fos signal quantification from **(J)**. Three independent experiments are shown. Bars represent the mean $\pm$ SD ($N = 3$). **** $P < 0.0001$ (one-way ANOVA with Tukey's post hoc test). **(L)** *(left)* Diagram of scFv(c-Fos) variants with reduced and increased binding affinities toward recombinant c-Fos, obtained through specific point mutations (T206K and T206E, respectively; see Methods for a detailed screening process). *(right)* Dose-response curves displaying the binding affinities of recombinant c-Fos-targeting scFvs for GST-tagged c-Fos. Immobilized scFvs were incubated with four-fold serial dilutions of GST-c-Fos (5000 – 0.32 nM) in an ELISA format. EC50 values were calculated from four-parameter-logistics equations (GraphPad Prism, ver. 10.4.1). The mean values were determined from three independent experiments, with error bars representing standard deviation ($N = 3$ each group). **(M)** *(left)* Schematic illustration of the experimental design assessing ternary complex stability among the Protea-Tac degrader, target substrate (c-Fos), and the 26S proteasome as a function of the tAb module's binding affinity. *(right)* HEK293T cells stably expressing biotin- and hexahistidine-tagged PSMB2 (PSMB2-HTBH) were transiently transfected with either control LacZ^{V5}, low-affinity scFv(c-Fos, T206K)-ADRM1, or high-affinity scFv(c-Fos, T206E)-ADRM1 constructs. Following streptavidin pull-down under native conditions, c-Fos and various proteasome subunits were analyzed by SDS-PAGE/IB with the indicated antibodies.

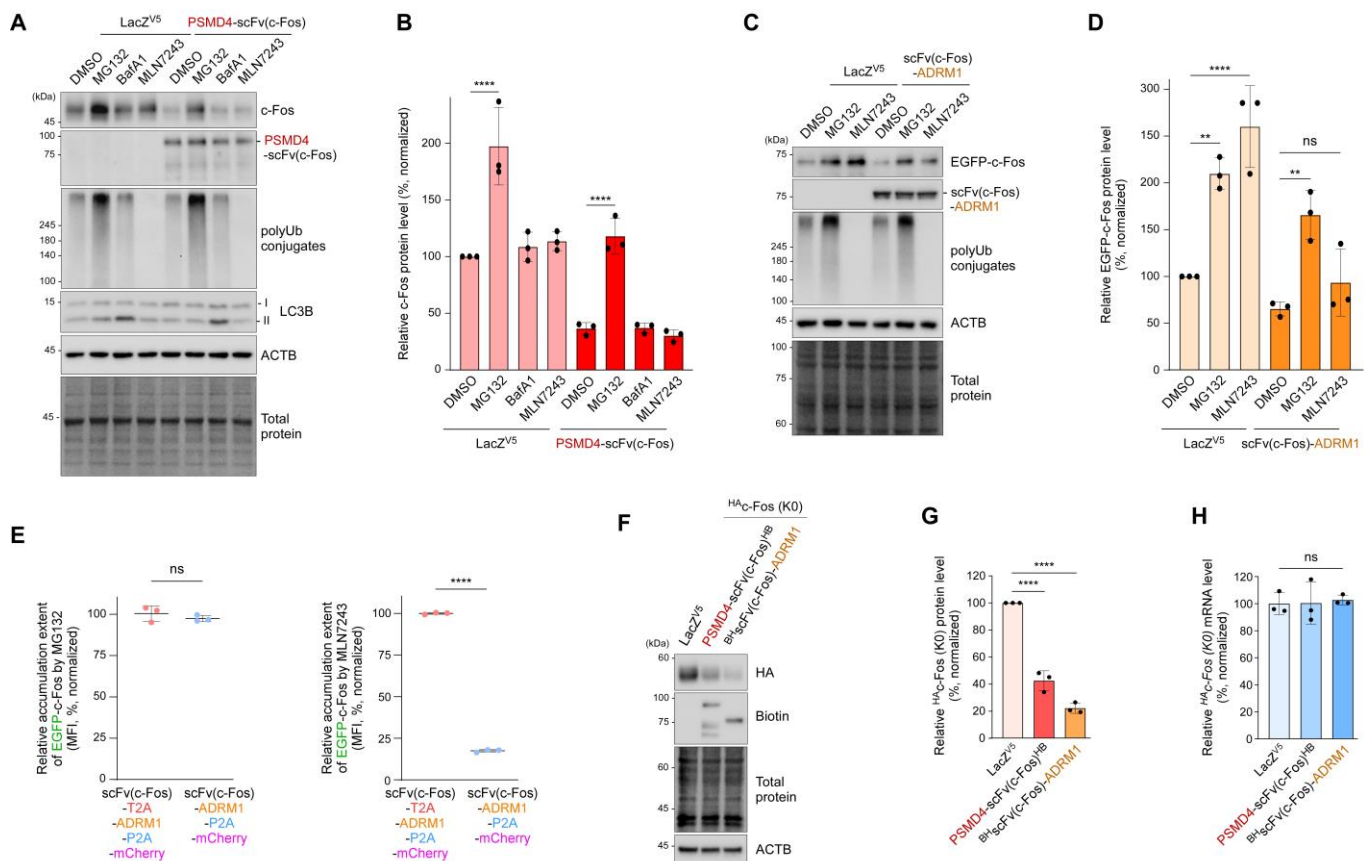


Fig. S3. The Protea-Tac degraders induce Ub-independent and proteasome-mediated TPD. Related to Figure 3.

(A) The PSMD4-scFv(c-Fos)-mediated c-Fos degradation assay was performed with DMSO, the proteasome inhibitor MG132 (10 μ M), the autophagic flux inhibitor bafilomycin A1 (BafA1; 200 nM), or the E1 Ub-activating enzyme inhibitor MLN7243 (1 μ M) for 6 h to determine pathway dependence. (B) Quantification of the steady-state levels of c-Fos from (A), normalized to total protein. Bars represent the mean $\pm$ SD (*N* = 3). *****P* < 0.0001 (one-way ANOVA with Tukey's post hoc test). (C) Similar to (A), but here the scFv(c-Fos)-ADRM1 was tested in the presence of MG132 (10 μ M for 6 h) or MLN7243 (1 μ M for 6 h) to evaluate its TPD activity toward EGFP-c-Fos. (D) Quantification of (C), normalized to total protein. Bars represent the mean $\pm$ SD (*N* = 3). ***P* < 0.01, *****P* < 0.0001; ns, not significant (one-way ANOVA with Tukey's post hoc test). (E) HEK293T cells stably expressing EGFP-c-Fos were transfected with the tricistronic control construct (red) or the bicistronic Protea-Tac construct (blue). After DMSO or MG132 (10 μ M for 6 h) or MLN7243 (1 μ M for 6 h) treatment, mean MFI value from EGFP-c-Fos in mCherry-positive cells were determined by flow cytometry. Extent of MFI value accumulation compared to DMSO group are shown (left: MG132, right: MLN7243). Bars indicate the mean $\pm$ SD (*N* = 3). *****P* < 0.0001; ns, not significant (two-tailed Student's *t* test). (F) A549 cells were co-transfected with lysine-less ^{HA}c-Fos (K0) and either LacZ^{V5} or the indicated anti-c-Fos Protea-Tac constructs for 64 h. WCLs were analyzed by SDS-PAGE followed by IB and total protein staining.

Representative IB results from three independent experiments are shown. **(G)** Quantification of ^{HA}c-Fos (K0) levels normalized to total protein signals. Bars represent the mean $\pm$ SD ($N = 3$). **** $P < 0.0001$ (one-way ANOVA with Tukey's post hoc test). **(H)** As in (F) except total RNA was extracted for quantitative RT-PCR using ^{HA}c-Fos (K0) and GAPDH (normalization control) primers. Values represent the mean $\pm$ SD of three independent experiments ($N = 3$). ns, not significant (one-way ANOVA with Tukey's post hoc test).

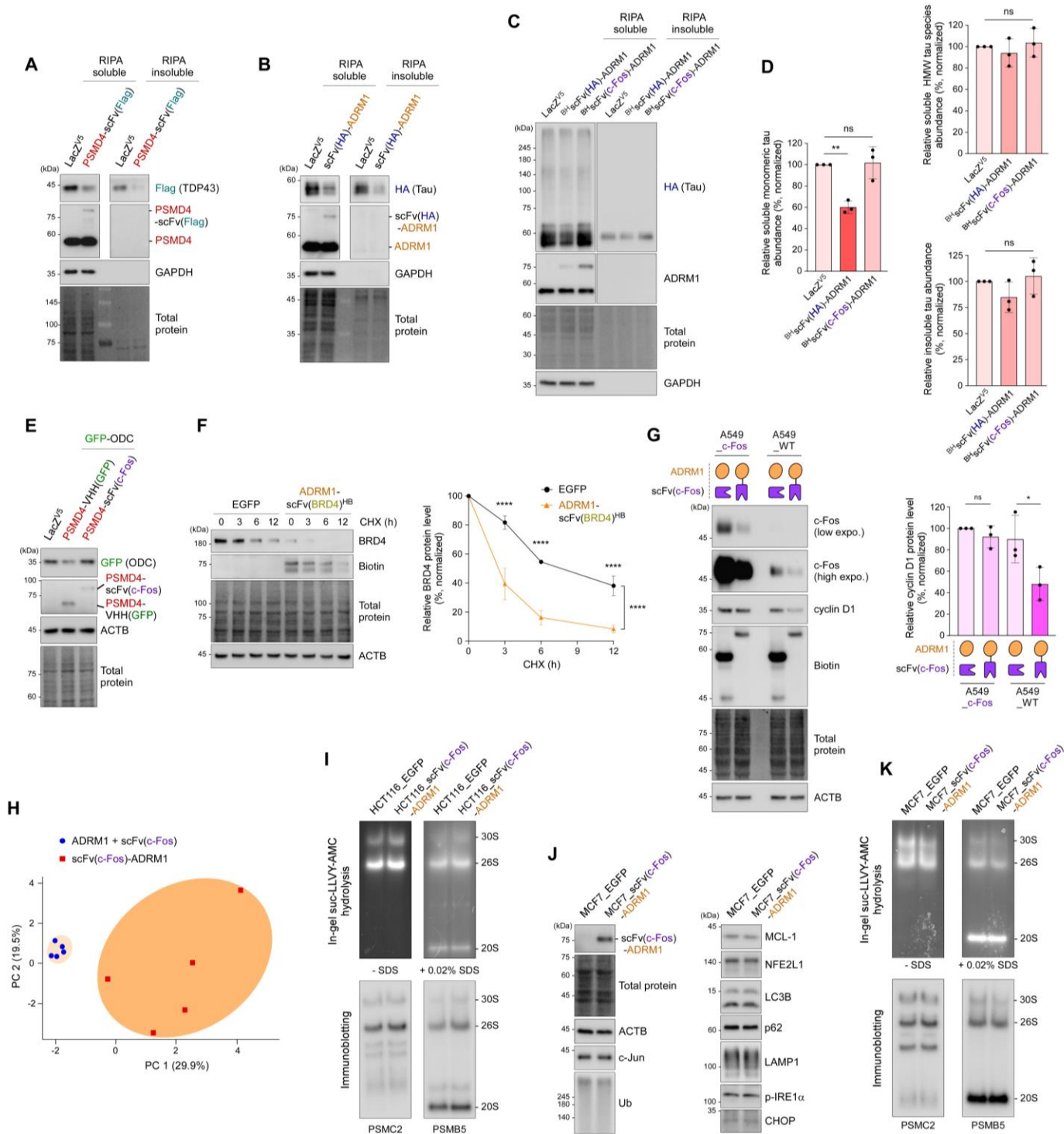


Fig. S4. The Protea-Tac chimeras allow for specific TPD of diverse proteins without affecting the global proteome. Related to Figure 4.

(A) A549 cells were co-transfected with $\text{Flag}^{\text{TDP43}}$ (target protein; 0.5 μg) and either PSMD4-scFv(Flag) (Flag degrader; 1 μg) or LacZ^{V5} (transfection control; 1 μg) for 72 h. WCLs were

separated into RIPA-soluble and -insoluble (pellet) fractions and examined using SDS-PAGE/IB and total protein staining. **(B)** Same as **(A)**, except ^{HA}Tau and scFv(HA)-ADRM1 were used as a target and degrader, respectively. **(C)** HEK293 cells stably expressing HA-tau(0N4R) were transfected with either LacZ^{V5} or ^{BH}scFv(HA)-ADRM1 or ^{BH}scFv(c-Fos)-ADRM1 for 64 h. Cells were lysed in RIPA buffer, whole cell lysates were separated into supernatant and pellet fractions and subjected to non-reducing (supernatants) and reducing (pellets) SDS-PAGE/IB. Representative IB results from three independent experiments are shown. **(D)** Left: quantification of RIPA-soluble, monomeric HA-tau(0N4R) levels normalized to total protein signals. *Top right*: levels of soluble, high-molecular-weight HA-tau(0N4R) species were quantified, normalized to total protein signals. *Bottom right*: RIPA-insoluble HA-tau(0N4R) levels were quantified, normalized to total protein signals. Bars represent the mean $\pm$ SD ($N = 3$). $**P < 0.01$, ns, not significant (one-way ANOVA with Tukey's post hoc test). **(E)** Specificity of Protea-Tac was assessed using mismatched target-degrader pairs. **(F)** A549 cells were treated with EGFP or anti-BRD4 Protea-Tac (ADRM1-scFv(BRD4)^{HB}) encoding lentivirus for 72 h and chase experiments were performed at 3, 6, and 12 h following treatment with 80 μ g/mL cycloheximide (CHX) at time zero. BRD4 signals were normalized to the values at 0 h chase (set as 100%) of each group and to total protein levels. Average percentages of remaining BRD4 (mean $\pm$ SD) from three independent experiments ($N = 3$, $****P < 0.0001$ from two-way ANOVA followed by Sidak's post hoc test). **(G)** A549 cells stably expressing c-Fos and A549 WT cells were transfected with either ADRM1^{HB} plus scFv(c-Fos)^{HB} or ^{BH}scFv(c-Fos)-ADRM1, for 64 h. WCLs were analyzed by SDS-PAGE followed by IB. *Left*: Representative IB results from three independent experiments are shown. *Right*: Quantification of cyclin D1 levels normalized to total protein signals. Bars represent the mean $\pm$ SD ($N = 3$). ns, not significant, $*P < 0.05$ (one-way ANOVA with Tukey's post hoc test). **(H)** Principal component analysis (PCA) plots depicting replicates from the control ($N = 5$; blue circles) and Protea-Tac-treated groups ($N = 5$; red rectangles). **(I)** Analysis of 26S proteasomes in HCT116 cells stably expressing EGFP or scFv(c-Fos)-ADRM1 using native PAGE, followed by in-gel suc-LLVY-AMC hydrolysis visualization (*top*) and subsequent IB with antibodies against the CP (PSMB5) and the RP (PSMC2) (*bottom*). 20S proteasomes were activated by 0.02% SDS. **(J)** Levels of various endogenous proteins were compared in MCF7_EGFP and MCF7_scFv(c-Fos)-ADRM1 stable cell lines by SDS-PAGE/IB with the appropriate antibodies. **(K)** As in **(I)**, except that MCF7_EGFP and MCF7_scFv(c-Fos)-ADRM1 stable cell lines were analyzed.

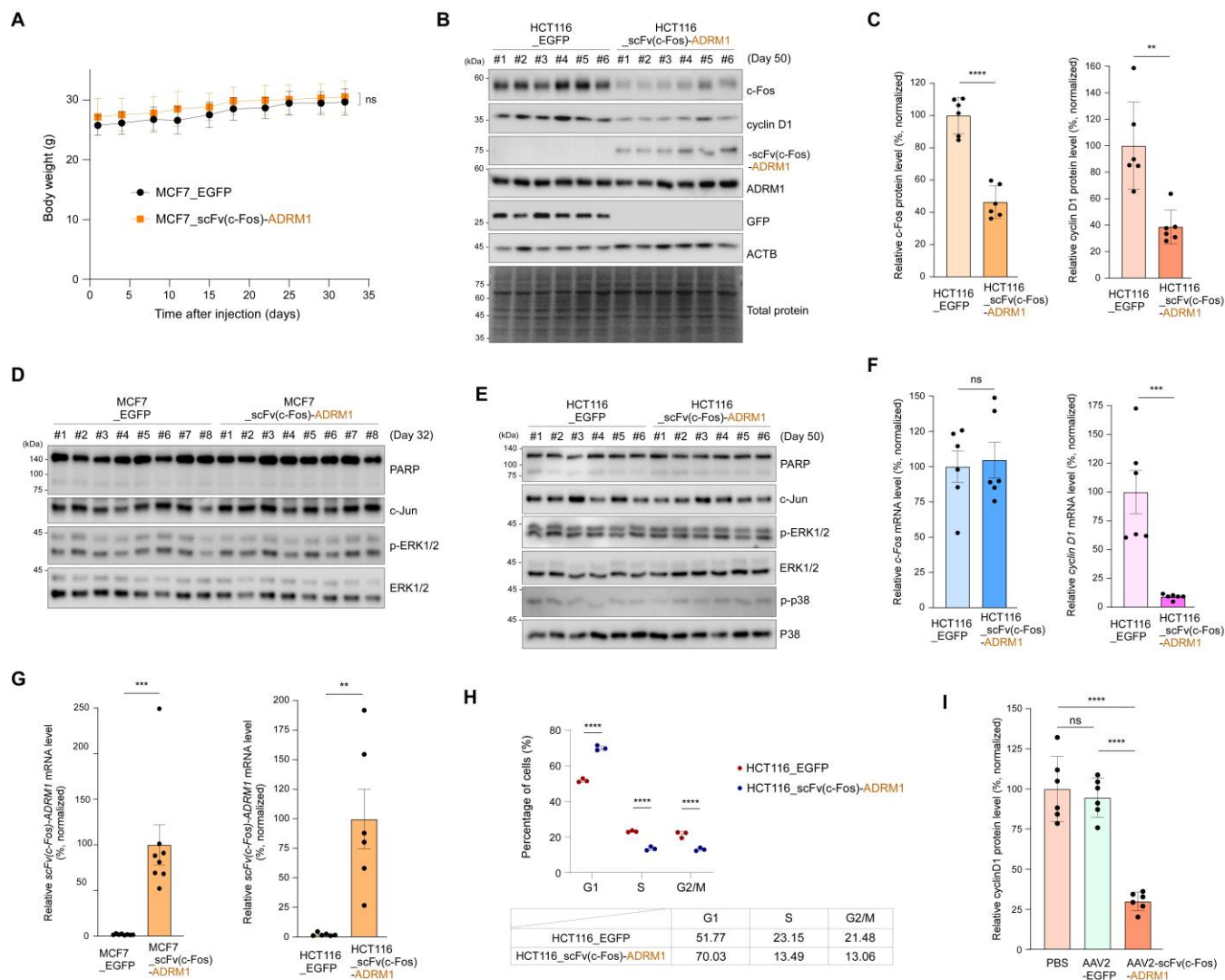


Fig. S5. Protea-Tac-induced c-Fos degradation leads to anti-tumor efficacy *in vivo*. Related to Figure 5.

(A) Body weight changes of NOD/SCID/IL-2 γ -receptor null (NSG) mice (male, six-week-old) xenografted with MCF7 cells stably overexpressing either EGFP or scFv(c-Fos)-ADRM1 ($N = 8$ each) throughout the 32-day experiment period. (B) Whole lysates of HCT116 xenograft tumors were subjected to SDS-PAGE/IB using the indicated antibodies. (C) Quantification of c-Fos and cyclin D1 protein levels in (B), normalized to total protein signals. Bars represent the mean $\pm$ SD ($N = 6$). $**P < 0.01$ and $****P < 0.0001$ (two-tailed Student's t-test). (D and E) IB analysis of xenograft tumor lysates from MCF7 (D) and HCT116 (E) stable cells, probed for apoptosis markers and upstream regulators of c-Fos. (F) qRT-PCR analysis of *c-Fos* and *cyclin D1* transcript levels in HCT116 xenograft tumors ($N = 6$). Quantitative normalization was performed using *GAPDH* mRNA expression. (G) As in (F), except that mRNA levels of *scFv(c-Fos)-ADRM1* were determined in biological triplicates. (H) Cell cycle distribution in the HCT116 stable cell lines used for xenograft assays, analyzed by flow cytometry. DNA contents were quantified using DAPI

staining. **** $P < 0.0001$ between EGFP- and Protea-Tac-expressing stable cell lines, two-way ANOVA followed by Bonferroni's multiple-comparisons test ($N = 3$ each). (I) Quantification of cyclin D1 protein levels from the IB analysis in Fig. 5J. Data are shown as mean $\pm$ SD ($N = 6$). **** $P < 0.0001$ (one-way ANOVA with Tukey's post hoc test). ns, not significant.

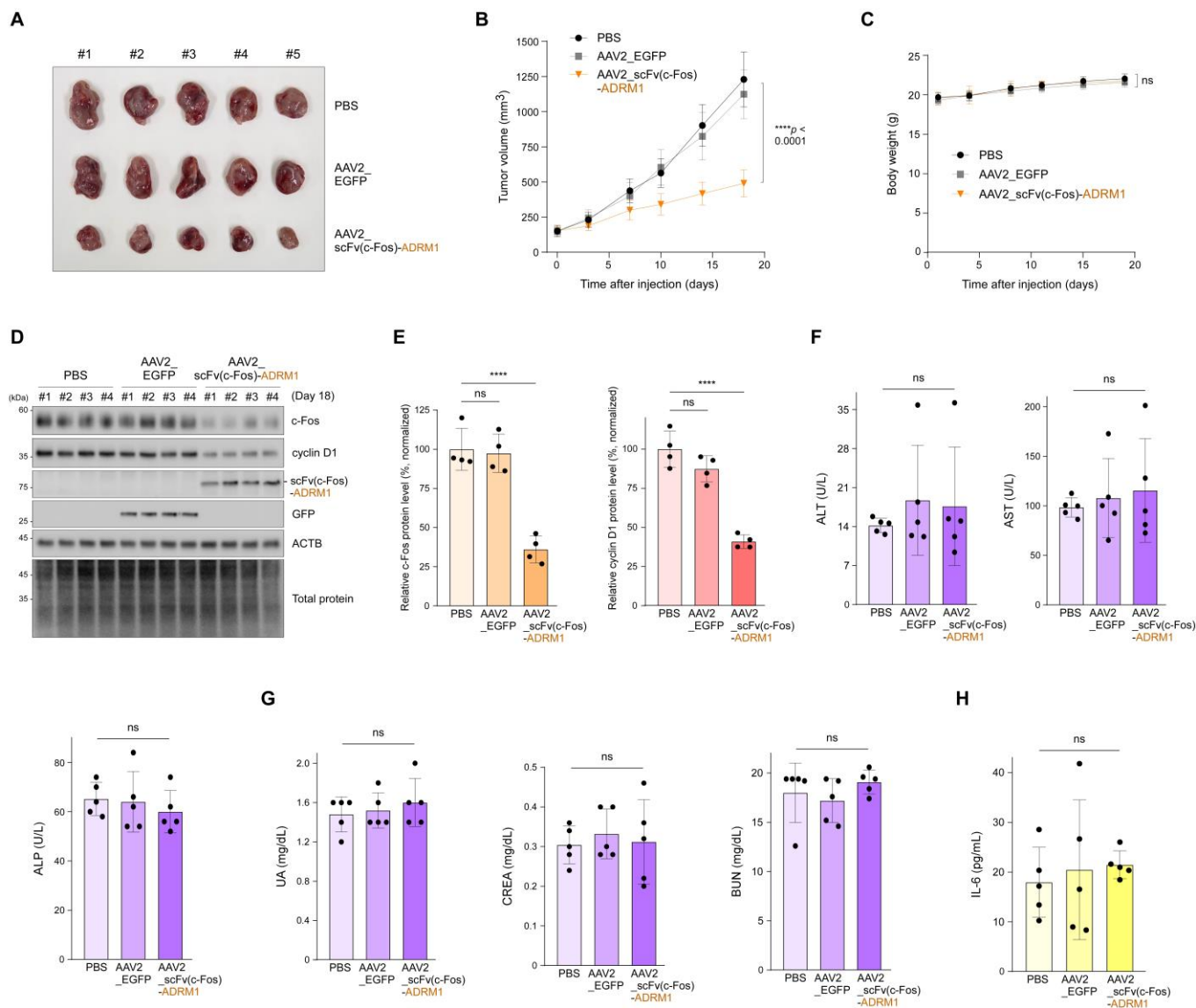


Fig. S6. AAV2-mediated Protea-Tac expression suppresses tumor growth in an immunocompetent syngeneic tumor model. Related to Figure 5.

(A) CT26 cells were subcutaneously inoculated into five-week-old female BALB/cAnNCrl mice. When tumors reached approximately 150 mm³, mice were intratumorally injected with PBS or AAV2 encoding EGFP or scFv(c-Fos)-ADRM1 (4×10^9 vg each) once weekly for a total of three injections. Representative images of tumors harvested on day 18 after the first injection are shown. (B) Tumor volumes were measured at the indicated time points for 18 days. Data are presented as means ± SEM (N = 5 per group). ****P < 0.0001 (two-way ANOVA followed by Tukey's post hoc test). (C) Body weight changes during the experiment. Data are presented as means ± SEM (N = 5 per group). ns, not significant (two-way ANOVA followed by Tukey's post hoc test). (D) IB analysis of whole-tumor lysates with the indicated antibodies. (E) Quantification of c-Fos (left) and cyclin D1 (right) protein levels in (D), normalized to total protein signals. Bars represent the

mean $\pm$ SD ($N = 4$). ns, not significant; **** $P < 0.0001$ (one-way ANOVA followed by Tukey's post hoc test). **(F and G)** Serum samples collected at the experimental endpoint were analyzed for alanine aminotransferase (ALT; F), aspartate aminotransferase (AST; F), alkaline phosphatase (ALP; F), uric acid (UA; G), creatinine (CREA; G), and blood urea nitrogen (BUN; G) levels. Bars represent the mean $\pm$ SD ($N = 5$). ns, not significant (one-way ANOVA followed by Tukey's post hoc test). **(H)** Serum IL-6 concentrations were measured at the experimental endpoint by ELISA. Bars represent the mean $\pm$ SD ($N = 5$). ns, not significant (one-way ANOVA followed by Tukey's post hoc test).

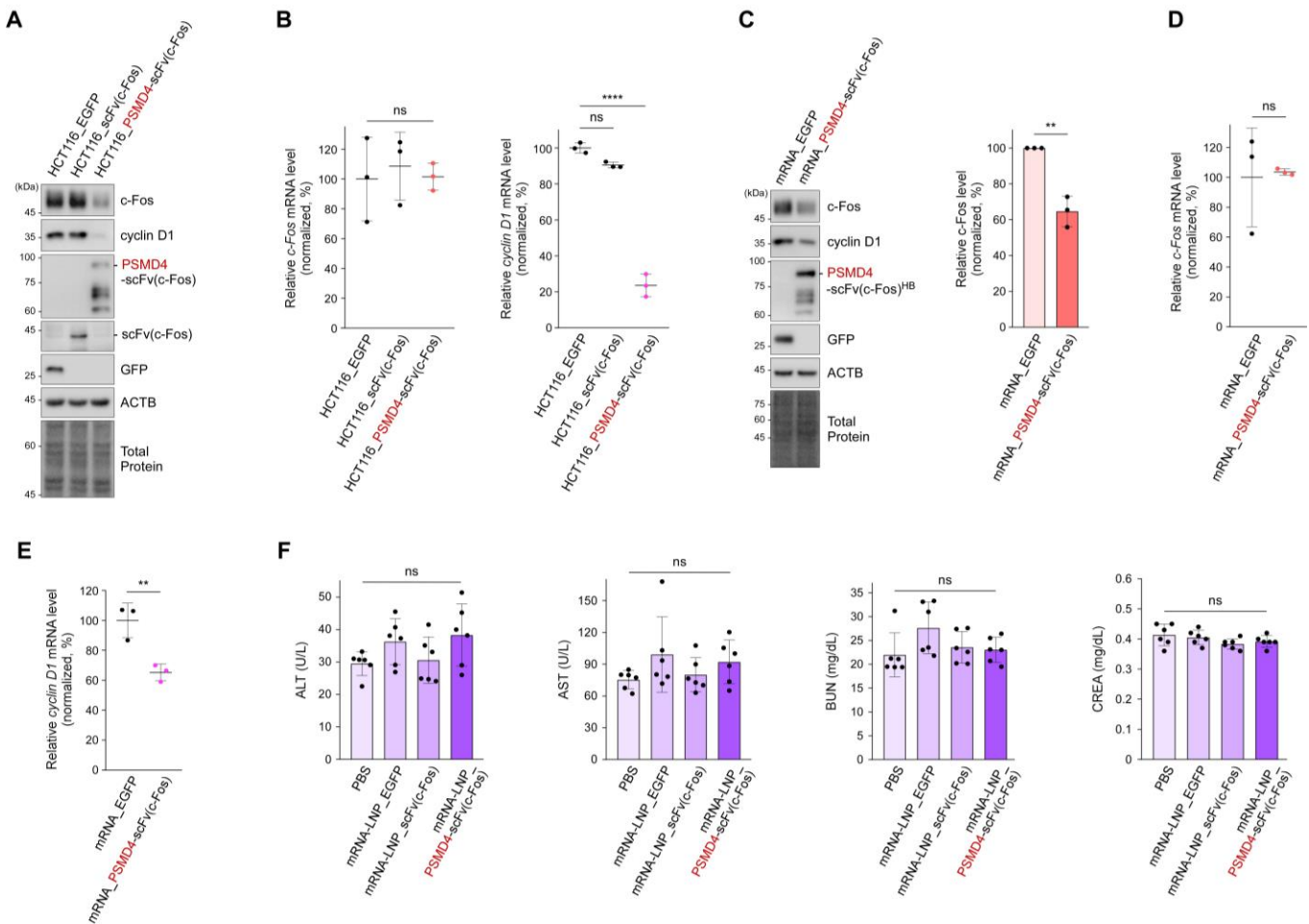


Fig. S7. *In vitro* validation and *in vivo* serum biochemical analysis of mRNA-mediated delivery of PSMD4-based Protea-Tac. Related to Figure 6.

(A) Endogenous c-Fos and cyclin D1 protein levels were compared in HCT116 cells stably expressing EGFP, scFv(c-Fos)^{HB} or PSMD4-scFv(c-Fos)^{HB}. Single-cell clones with the highest expression level were selected for analysis. (B) As in (A) except that mRNA levels of endogenous *c-Fos* (left) and *cyclin D1* (right) were evaluated by qRT-PCR and normalized to GAPDH. ns, not significant; ****P < 0.0001 (one-way ANOVA with Tukey's post hoc test). (C) A549 cells were transfected with EGFP or PSMD4-scFv(c-Fos)^{HB} encoding mRNA for 36 h. WCLs were analyzed by SDS-PAGE and IB. Left: Representative IB results from three independent experiments are shown. Right: Quantification of c-Fos levels normalized to total protein signals. Bars represent the mean ± SD (N = 3). **P < 0.01 (two-tailed Student's t-test). (D) As in (C) except that mRNA levels of endogenous *c-Fos* were evaluated by qRT-PCR and normalized to GAPDH. Values are presented as means ± SD. ns, not significant (N = 3, two-tailed Student's t-test). (E) As in (C) except that mRNA levels of endogenous *cyclin D1* were evaluated by qRT-PCR and normalized to GAPDH. Values are presented as means ± SD. **P < 0.01 (N = 3, two-tailed Student's t-test). (F) Serum samples collected from the mice used in the xenograft experiment shown in Figure 6 at the experimental endpoint were analyzed for alanine aminotransferase (ALT), aspartate

aminotransferase (AST), blood urea nitrogen (BUN), and creatinine (CREA) levels. Bars represent the mean $\pm$ SD ($N = 6$). ns, not significant (one-way ANOVA followed by Tukey's post hoc test).

Table S1. Coding sequences of chimeras.

Color code : **Proteasome subunit** – **Linker** – **scFv/VHH**

PSMD4- scFv(c-Fos)	ATGGTGTTGGAAAGCACTATGGTGTGTGTGGACAACAGTGAGTATATGCGGAATGGAGACTTCTTACCCACAGGC TGCAGGCCCAGCAGGATGCTGTCAACATAGTTTGTTCATTCAAAGACCCGACGCAACCCTGAGAACAACTGGGGCC TTATCACACTGGCTAATGACTGTGAAGTGCTGACCACACTCACCCAGACACTGGCCGTATCTGTCCAAGCTACA TACTGTCCAACCCAAAGGGCAAGATCACCTTCTGCACGGGCATCCGCGTGGCCCATCTGGCTCTGAAGCACCGACA AGGCAAGAATCACAAAGATGCGCATCATTTGCCCTTTGTGGGAAGCCAGTGAGGAGACAATGAGAAGGATCTGGTGAA ACTGGCTAAACGCCTCAAGAAGGAGAAAAGTAAATGTTGACATTATCAATTTTGGGGAAGAGGAGGTGAACACAGA AAAGCTGACAGCCTTTGTAACACGTTGAATGGCAAAGATGGAACCGGTTCTCATCTGGTGACAGTGCCTCCTGG GCCCAGTTTGGCTGATGCTCTCATCAGTTCTCCGATTTTGGCTGGTGAAGGTGGTGCCATGCTGGGTCTTGGTGCC AGTGACTTTGAATTTGGAGTAGATCCCAGTGCTGATCCTGAGCTGGCCTTGGCCCTTCGTGTATCTATGGAAGAGC AGCGGCAGCGGCAGGAGGAGGAGGCCCGGGCAGCTGCAGCTTCTGCTGCTGAGGCCGGGATTGCTACGACT GGGACTGAAGACTCAGACGATGCCCTGCTGAAGATGACCATCAGCCAGCAAGAGTTTGGCCGCACTGGGCTTCCT GACCTAAGCAGTATGACTGAGGAAGAGCAGATTGCTATGCCATGCAAGTGCATGCCGTGAGGAGCGAGATTGGC CAGGCGGAATCAGCAGACATTGATGCCAGCTCAGCTATGGACACATCTGAGCCAGCCAAGGAGGAGGATGATTAC GACGTGATGCAGGACCCGAGTTCCTTCAGAGTGTCTAGAGAACCCTCCAGGTGTGGATCCCAACAATGAAGCC ATTGAAATGCTATGGGCTCCCTGGCCTCCAGGCCACCAAGGACGGCAAGAAGGACAAGAAGGAGGAAGACAA GAAGCGACCCAAAGCTTGGTACCGAGCTCGGATCCATGGAAGTGCAGCTGCAGCAGAGAGCGGAGCGAGTTGGC GGAGCGGGGCTCTGTGAAACTGTCTTGACCCGCTTCTGGCTTCAACATCAAGGACTACTACATGCACTGGGTCA AGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGATGGACCCCAAGAAGCGCGATACCGAGTACGCCCTTA AGTTCCAGGGCAAGGCCACCATGACCGCCGACACCCAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAA GCCAGCACACAGCCGTGTACTACTGCCACGACGCCATCACCACGCCACAGAACTTCGACGTGTGGGGCGCCG GCACAACCGTGACCGTGTCTTCTGGCGGGGAGGCGAGCGCGGAGGCGGCGAGCGCGGCGGCGGCGGAGGAGG CGGCTCTGATGTGGTGATGACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGACCAAGGCCAGCATCAGCTG TAGAAGCTCTCAGAGCCTGGTGACAGCAACGGCAACACCTACCTGCACTGGTACCTGCAAAAGCCCGGCCAGTC TCCAAAGCTGCTGATGTACAAGGTGTCCAATAGATTACGGGTGTTCTTGATAGATTAGCGGCAGCGGGAGCGGA ACCGATTCACCTGAAGATCAGCAGAGTGGAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACAC GTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATCAAG
scFv(c-Fos)- ADRM1	ATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCGGAGCGGGGCTCTGTGAAACTGTCTTGCACCGC TTCTGGCTTCAACATCAAGGACTACTACATGCACTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGG CTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACAC CAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGC CATCACACGCCACCCAGAACTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGGCGAG GCAGCGGGGAGGCGGCGAGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGG AGCCTGCTGTGTCCCTGGGCGACACGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACCGG AACACCTACCTGCACTGGTACCTGCAAAAGCCCGGCCAGTCTCCAAAGCTGCTGATGTACAAGGTGTCCAATAGAT TCAGCGGTGTTCTTGATAGATTAGCGGCAGCGGGAGCGGAACCGACTTCACCTGAAGATCAGCAGAGTGGAAG CCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCCTTACCTTCGGCTCCGGGCACCAAGCTCG AGATCAAGGAATTCATGACGACCTCAGGCGCGCTCTTTCCAAGCCTGGTGCCAGGCTCTCGGGGCGCCTCCAACA AGTACTTGGTGGAGTTTCGGGGCGGAAAGATGTCTCTGAGGGGACACCGTGACTCCGGGAGCGGAAAGGG CTGGTGATATTACAGCAGACGGACGACTCGCTATTCACTTCTGCTGGAAGGACAGGACGTCGGGAACGTGGAA GACGACTTGATCATCTTCCCTGACGACTGTGAGTCAAGCGGGTGCCGAGTGCCCCAGCGGGAGGGTCTACGTG CTGAAGTTCAAGGACGGGTCCAAGCGGCTTTCTTCTGGATGCAGGAACCAAGACAGACAGGATGAGGAGCAT TGCCGCGAGACCGCGGATGAGATCCAGAATACCCTGACCTCGCCCAAGTTCCAGCAGGCCGCTGGGCGGCGGCA CGAACTCTCTGCGCTAGGCGGTGAGGGTGGCCTGCAGAGCCTGCTGGGAAACATGAGCCACAGCCAGCTCATGCA GCTCATCGGACAGCCGGCCTTGGAGGACTGGGTGGGCTGGGGGCCCTGACTGGACCTGGCCTGGCCAGCTTACT GGGGAGCAGTGGGCCTCCAGGGAGCAGCTCCTCCTCCAGCTCCCGAGCCAGTCGGCAGCGGTACCCCCGTCAT CCACCACCTCTTCCACCCGTGCCACCCAGCCCTTCTGCTCCAGCAGTGCCTCAGCAACTAGCCCCAGCCCCGC GCCCAGTTCGGGAATGGAGCCAGCACAGCAGCCAGCCGACCCAGCCATCCAGCTGAGCGACCTCCAGAGCA TCCTGGCCACGATGAACGTACCAGCCGGGCCAGCAGGCGGCCAGCAAGTGGAACCTGGCCAGTGTGCTGACGCCG GAGATAATGGCTCCCATCTCGCAACGCGGATGTCCAGGAGCGCCTGCTTCCCTACTTGCCATCTGGGGAGTTCG TGCCGCGAGACCGCGGATGAGATCCAGAATACCCTGACCTCGCCCAAGTTCCAGCAGGCCGCTGGGATGTTTACGG CAGCCTTGGCCTCGGGGAGCTGGGCCCCCTCATGTGCCAGTTCGGTCTGCCTGCAGAGGCTGTGGAGGCCGCCA ACAAGGGCGATGTGGAAGCGTTTGCCAAAGCCATGCAGAAACGCCAAGCCCCAGCAGAAAGAGGGCGACAC GAAGGACAAGAAGGACGAAGAGGAGGACATGAGCCTGGAC
ADRM1- scFv(c-Fos)	ATGACGACCTCAGGCGCGCTCTTTCCAAGCCTGGTGCCAGGCTCTCGGGGCGCCTCCAACAAGTACTTGGTGGAG TTTCGGGCGGAAAAGATGTCCCTGAAGGGGACACCGTGACTCCGGATAAGCGGAAAGGGCTGGTGTACATTCAG CAGACGGACGACTCGCTTATTCACTTCTGCTGGAAGGACAGGACGTCGGGAACGTGGAAGACGACTTGATCATC TTCCCTGACGACTGTGAGTTCAAGCGGGTGCCGAGTGCCCCAGCGGGAGGGTCTACGTGCTGAAGTTCAAGGCA GGGTCCAAGCGGCTTTTCTTCTGGATGCAGGAACCAAGACAGACAGGATGAGGAGCATTGCCGGAAGTCAAA CGAGTATCTGAACAACCCCGGATGCTGGGGCGCTGGGGGCCAGCGGAAGCAGCAACTCTCTGCGCT AGGCGGTGAGGGTGGCCTGCAGAGCCTGCTGGGAAACATGAGCCACAGCCAGCTCATGCAGCTCATCGGACAG CCGCCCTTGGAGGACTGGGTGGGCTGGGGGCCCTGACTGGACCTGGCCTGGCCAGCTTACTGGGGAGCAGTGGG CCTCCAGGGAGCAGCTCCTCCTCCAGCTCCCGAGCCAGTCGGCAGCGGTACCCCCGTCATCCACCACCTCTTCC ACCCGTGCCACCCAGCCCCCTTCTGCTCCAGCAGCTGCCTCAGCAACTAGCCCCAGCCCCGCGCCAGTTCGGG

	AATGGAGCCAGCACAGCAGCCAGCCCCAGCCCAGCCCATCCAGCTGAGCGACCTCCAGAGCATCCTGGCCACGATG AACGTACCAGCCGGGCCAGCAGGCGGCCAGCAAGTGGACCTGGCCAGTGTGCTGACGCCGGAGATAATGGCTCC CATCCTCGCCAACGCGGATGTCCAGGAGCGCCTGCTTCCCTACTTGGCATCTGGGGAGTGCCTGCCGCAGACCGCG GATGAGATCCAGAATACCCTGACCTCGCCCCAGTTCCAGCAGGCCCTGGGCATGTTTCAGCCGACGCTTGGCCTCGG GGCAGCTGGGCCCCCTCATGTGCCAGTTCGGTCTGCCTGCAGAGGCTGTGGAGGCCGCCAACAAAGGGCGATGTGG AAGCGTTTGCCAAAGCCATGCAGAACACGCCAAGCCCCAGCAGAAAGAGGGCGACACGAAGGACAAGAAGGA CGAAGAGGAGGACATGAGCCTGGACAAAGCTTGGTACCGAGCTCGGATCCATGGAAGTGCAGCTGCAGCAGAGCG GAGCCGAGCTGGTGCAGCGGGGCCCTCTGTGAAACTGTCTTGACCCGTTCTGGCTTCAACATCAAGGACTACT ACATGCACTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGATGGACCCCAAGAACGGCGAT ACCGAGTACGCCCTTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACACCAGCAGCAATACCGCCTACCTGCAG CTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCCATACCCACGCCACCAGAAAACCTTC GACGTGTGGGGCGCCGGCAACCCGTGACCGTGTCTTCTGGCGGGCAGGCAGCGCGGAGGCGGACGAGTGGCGG GCGCGCGCAGCGGAGGCGGCTCTGATGTGGTGAAGTACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGCGGACC AGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGCAACACCTACCTGCACTGGTACCTGC AAAAGCCCCGCCAGTCTCCAAAGCTGCTGATGTACAAGGTGTCCAATAGATTACGCGGTGTTCTGTATAGATTTAG CGCAGCGGGAGCGGAACCGACTTACCCTGAAGATCAGCAGAGTGAAGCGGAAGATCTGGGAGTGTACTTCT GCAGCGTCAACACACGTCCTTTCACCTTCGGCTCCGGCACCAAGCTCGAGATCAAG
scFv(c-Fos)- PSMD2	ATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCAGGAGCGGGGCCCTCTGTGAAACTGTCTTGACCCG TTCTGGCTTCAACATCAAGGACTACTACATGCCTGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGG CTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACAC CAGCAGCAATACCGCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGCGC CATCACACGCCACCAGAAAACCTTCGACGTGTGGGGCGCCGGCAACCCGTGACCGTGTCTTCTGGCGGCGGAG GCAGCGCGGAGGGCGGAGCGCGGGCGGCGGAGCGGAGCGGCTCTGATGTGGTGTGATGACCCAGACACCTCTG AGCCTGCCTGTGTCCCTGGGCGACAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGC AACACCTACCTGCACTGGTACCTGCAAAAGCCCGGCCAGTCTCCAAAGCTGCTGATGTACAAGGTGTCCAATAGAT TCAGCGGTGTTCTGTATAGATTTAGCGCGCAGCGGGAGCCGAACCGACTTACCCTGAAGATCAGCAGAGTGGAA CCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTCACCTTCGGCTCCGGCACCAAGCTCG AGATCAAGGAATTATGAGGAGGGAGCGCGGGACAAGGCGCCGGTGCAGCCCCAGCAGTCTCCAGCGGCGGGC CCCGGCGGCACGGACGAGAAGCCGAGCGGCAAGGAGCGCGGGATGCCGGGACAAGGACAAGAAGACAGGAG CTGTCTGAAGAGGATAAACAGCTTCAAGATGAAGTGGAGATGCTCGTGGAACGACTAGGGGAGAAGGATACATCC CTGTATCGACCAGCGCTGGAGGAATTGCGAAGGCAGATTCTGTTCTTCTACAACCTCCATGACTTCAGTGCCCAAGC CTCTCAAATTTCTGCGTCCACACTATGGCAAACCTGAAGGAAATCTATGAGAACATGGCCCTGGGGAGAATAAGCG TTTTGCTGCTGACATCATCTCCGTTTTGGCCATGACCATGAGTGGGGAGCGTGAAGTGCCTCAAGTATCGGCTAGTG GGCTCCCAGGAGGAATTGGCATATGGGGTCAAGATGTGTCAGGCATCTGGCAGGAGAAGTGGCTAAGGAGTGG CAGGAGCTGGATGACGCAGAGAAGGTCCAGCGGGAGCCTCTGCTCACTCTGGTGAAGGAAATCGTCCCCTATAAC ATGGCCACAATGCAGAGCATGAGGCTTGCGACCTGCTTATGGAATGAGCAGGTGGACATGCTGGAGAAGGAC ATTGATGAAAATGCATATGCAAAGGTCTGCCTTTATCTACCAAGTTGTGTGAATTACGTGCCTGAGCCTGAGAAGT AGCCCTATGCGTTGTGCCCTGGGTGTGTTCCGAAAGTTAGCCGCTTCCCTGAAGTCTGAGATTGGCATGTATG CTCAATGACATGGAGTTGGTAGAAGACATCTTCAAGCTCTGCAAGGATGTGGTAGTGAAGATGAGATTGGCATTCA TGCTAGGCCGGCATGGGGTGTCTGAGGCTGAGTGAAGATGTGAGGAGTATGAGGACCTGACAGAGATCATGT CCAATGTACAGCTCAACAGCAACTTCTTGGCCTTAGCTCGGGAGCTGGACATCATGGAGCCCAAGGTGCCTGATG ACATCTACAAAACCCACCTAGAGAACAACAGGTTTGGGGGAGTGGCTCTCAGGTGGACTCTGCCCGCATGAACC TGGCCTCCTCTTTTGTGAATGGCTTTGTGAATGCAGCTTTTGGCCAAAGACAAGCTGAACAGATGATGGCAACAA ATGGCTTTACAAGAACAAGGACCACGGAATGTTGAGTGCAGCTGCATCTCTTGGGATGATTCTGCTGTGGGATGTG GATGGTGGCCTCACCCAGATTGACAAGTACCTGTACTCCTCTGAGGACTACATTAAGTCAAGGAGCTCTTCTGCT GTGGCATAGTGAAGTCTGGGGTCCGGAATGAGTGTGACCCCTGCTCTGGCACTGCTCTCAGACTATGTTCTCCACAA CAGCAACACCATGAGACTTGGTTCCATCTTTGGCTAGGCTTGGCTTATGCTGGCTGAAGATGATGATGCTCTCA ACACTGCTGCTGCCTGTGATGGGAGATTCAAAGTCCAGCATGGAGGTGGCAGGTGTCACAGCTTTAGCCTGTGGA ATGATAGCAGTAGGGTCTGCAATGGAGATGTAACCTCCACTATCCTTCAGACCATCATGGAGAAGTCAGAGACTG AGCTCAAGGATACTTATGCTCGTTGGCTTCTCTTGGACTGGGTCTCAACCACCTGGGGAAGGGTGAGGCCATCGA GGCAATCCTGGCTGCACCTGGAGGTGTGTGCAGAGCCATTCCGAGTTTTTGCCAACACACTGGTGGATGTGTGCA TATGCAGGCTCTGGGAATGTGCTGAAGGTGCAGCAGCTGCTCCACATTTGTAGCGAACACTTTGACTCCAAAGAG AAGGAGGAAGACAAAGACAAGAAGGAAAAAGACAAGGACAAGAAGGAAGCCCCCTGCTGACATGGGAGCA CATCAGGGAGTGGCTGTTCTGGGGATTGCCCTTATTGCTATGGGGGAGGAGATTGGTGCAGAGATGGCATTACGAA CCTTTGGCCACTTGTGAGATATGGGGAGCCTACACTCCGAGGGCTGTACCTTTAGCACTGGCCCTCATCTCTGTT TCAAAATCCAGCACTCAACATCCTGGATACCCTAAGCAAAATCTCTCATGATGCTGATCCAGAAGTTTCCATAACTC CATTTTTGCCATGGGCATGGTGGGCAGTGGTACCAATAATGCCCCGTCTGGCTGCAATGCTGCGCCAGTTAGCTCAAT ATCATGCCAAGGACCCAAACAACCTTTCATGGTGCCTTGGCACAGGGCCTGACACATTTAGGGAAGGGCACCC TTACCCTCTGCCCCTACCACAGCGACCGGACGCTTATGAGCCAGGTGGCCGTGGCTGGACTGCTCACTGTGCTTGT CTCTTTCCCTGGATGTTGGAACATTATTCTAGGCAAAATCACACTATGTATTGTATGGCTGGCTGCCATGCAGCC CCGAATGCTGGTTACGTTTGTGATGAGGAGCTGCGGCCATTGCCAGTGTCTGTCCGTGTGGGCCAGGCAGTGGATGTG GTGGGCCAGGCTGGCAAGCCGAAGACTATCACAGGGTTCAGACGCATACAACCCCAAGTGTGTTGGCCACGGG GAACGGGCAGAATTGGCCACTGAGGAGTTTCTCTGTTACCCCCATTCTGGAAGGTTTTGTTATCCTTCGGAAGA ACCCCAATTATGATCTC
scFv(c-Fos)- (GGGS) ₃ -	ATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCAGGAGCGGGGCCCTCTGTGAAACTGTCTTGACCCG TTCTGGCTTCAACATCAAGGACTACTACATGCCTGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGG CTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACAC CAGCAGCAATACCGCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGC

	ATGGAGCCCAAGGTGCCTGATGACATCTACAAAACCCACCTAGAGAACAACAGGTTTGGGGGCAGTGGCTCTCAG GTGGACTCTGCCCGCATGAACCTGGCCTCCTCTTTTGTGAATGGCTTTGTGAATGCAGCTTTTGGCCAAGACAAGC TGCTAACAGATGATGGCAACAAATGGCTTTACAAGAACAAGGACCACGGAATGTTGAGTGCAGCTGCATCTCTTG GGATGATTCTGCTGTGGGATGTGGATGGTGGCCTCACCCAGATTGACAAGTACCTGTACTCTCTGAGGACTACATT AAGTCAGGAGCTCTTCTTGCCTGTGGCATAGTGAACTCTGGGGTCCGGAATGAGTGTGACCCGTGCTCTGGCACTGC TCTCAGACTATGTTCTCCACAACAGCAACACCATGAGACTTGGTTCCATCTTTGGGCTAGGCTTGCTTATGCTGGC TCAAATCGTGAAGATGTCCTAACACTGCTGCTGCCTGTGATGGGAGATTCAAAGTCCAGCATGGAGGTGGCAGGT GTCACAGCTTTAGCCTGTGGAATGATAGCAGTAGGGTCCTGCAATGGAGATGTAACCTCCACTATCCTTCAGACCAT CATGGAGAAGTCAGAGACTGAGCTCAAGGATACTTATGCTCGTTGGCTTCTCTTGGACTGGGTCTCAACCACCTG GGGAAGGGTGAGGCCATCGAGGCAATCCTGGCTGCACTGGAGGTTGTGTGAGAGCCATTCCGCGAGTTTGGCAAC ACACTGGTGGATGTGTGTCATATGCAGGCTCTGGGAATGTGCTGAAGGTGCAGCAGCTGCTCCACATTTGTAGCG AACACTTTGACTCCAAAGAGAAGGAGGAAGACAAGACAAGGAAAAGAAAGACAAGGACAAGAAGGAAG CCCTGCTGACATGGGAGCACATCAGGGAGTGGCTGTTCTGGGGATTGCCCTTATGGCTTGGGGGAGGAGATTGG TGCAGAGATGGCATTACGAACCTTTGGCCACTTGCTGAGATATGGGGAGCCTACACTCCGGAGGGCTGTACCTTTA GCACTGGCCCTCATCTCTGTTTCAAATCCACGACTCAACATCCTGGATACCTTAAGCAAATTTCTCATGATGCTGA TCCAGAAGTTTCTATAACTCCATTTTGGCATGGGCTAGGTGGGCAGTGGTACCAATAATGCCCCGTCTGGCTGCAA TGCTGCGCCAGTTAGCTCAATATGCTGCAAGGACCCAAACCTCTTCATGGTGGCTTGGGAGGAGCTGAC ACATTTAGGGAAGGGCACCTTACCCTCTGCCCTACCACAGCGACCGGCAGCTTATGAGCCAGGTGGCCGTGGC TGGACTGCTCACTGTGCTTGTCTCTTCTGGATGTTCGAAACATTATTCTAGGCAAATCACACTATGTATTGTATGG GCTGGTGGCTGCCATGCAGCCCCGAATGCTGGTTACGTTTGTAGAGGAGCTGCGGCCATTGCCAGTGTCTGTCCGT GTGGCCAGGCAGTGGATGTGGTGGGCCAGGCTGCGGAAGCAGGACTATCAGAGGTTCCGACAGGCTACACAC CCCAGTGTTGTTGGCCACGGGGAACGGGCAGAATTGGCCACTGAGGAGTTTCTTCTGTATACCCCATTTCTGGAA GGTTTGTATCCTTCGGAAGAACCCCAATTATGATCTC
PSMA4- scFv(c-Fos)	ATGTCTCGAAGATATGACTCCAGGACCACTATATTTTCTCCAGAAGGTGCGTTATACCAAGTTGAATATGCCATGGA AGCTATTGGACATGCAGGCACCTGTTTGGGAATTTTAGCAAATGATGGTGTGTTTGTCTGCAGCAGAGAGACGCAAC ATCCACAAGCTTCTTGATGAAGTCTTTTCTTGAAAAAATTATAAACTCAATGAGGACATGGCTTGCAGTGTGGC AGGCATAACTTCTGATGCTAATGTTCTGACTAATGAACTAAGGCTCATTGCTCAAAGGTATTTATTACAGTATCAGGA GCCAATACCTTGTGAGCAGTTGGTTACAGCGCTGTGTGATATCAAACAAGCTTATACACAATTTGGAGGAAAACGT CCCTTTGGTGTTCATTGCTGTACATTGGCTGGGATAAGCACTATGGCTTTCAGCTCTATCAGAGTGACCCTAGTGG AAATTACGGGGGATGGAAGGCCACATGCATTGGAATAATAGCGCTGCAGCTGTGTCAATGTTGAAACAAGACTAT AAAGAAGGAGAAATGACCTTGAAGTCAGCACTTGCTTTAGCTATCAAAGTACTAAATAAGACCATTGGATGTTAGTA AACTCTCTGCTGAAAAAGTGGAAATTGCAACACTAACAAGAGAGAATGGAAGACAGTAATCAGAGTTCTCAAA CAAAAAGAAGTGAGCAGTTGATCAAAAAACATGAGGAAGAAGAAGCCAAAGCTGAGCGTGAGAAGAAAAGAAA AAGAACAGAAAGAAAAGGATAAACGATCCATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGGCGGAG CGGGGCTCTGTGAAACTGTCTTGACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCAGTGGGTCAAGCAG AGACCTGAGCAGGGCCTGGAATGGATCGGCTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTC CAGGGCAAGGCCACCATGACCGCCGACACCAGCAGCAATACCGCTACCTGCAGCTGAGCAGCCTTACAAGCGAC GACACAGCCGTGTACTACTGCCACGACGCCATCACCACGCCACCAGAACTTCGACGTGTGGGGCGCCGCGACA ACCGTGACCGTGTCTTCTGGCGCGGAGGCGAGCGCGGCGGCGGCGAGCGCGGCGGCGGCGGCGGCT CTGATGTGGTGTGATGACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGACCAGGCCAGCATCAGCTGTAGAA GCTCTCAGAGCCTGGTGCACAGCAACGGCAACACCTACCTGCAGTGGTACCTGCAAAAGCCCGGCCAGTCTCAA AGCTGCTGATGTACAAGGTGTCCAATAGATTACGCGGTGTTCTGTATAGATTTAGCGGCAGCGGGAGCGGAACCGA CTCACCCCTGAAGATCAGCAGAGTGGAAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCC TTTACCTTCGGCTCCGGCACCAAGCTCGAGATCAAG
PSMA4- (GGGGS) ₃ - scFv(c-Fos)	ATGTCTCGAAGATATGACTCCAGGACCACTATATTTTCTCCAGAAGGTGCGTTATACCAAGTTGAATATGCCATGGA AGCTATTGGACATGCAGGCACCTGTTTGGGAATTTTAGCAAATGATGGTGTGTTTGTCTGCAGCAGAGAGACGCAAC ATCCACAAGCTTCTTGATGAAGTCTTTTCTTGAAAAAATTATAAACTCAATGAGGACATGGCTTGCAGTGTGGC AGGCATAACTTCTGATGCTAATGTTCTGACTAATGAACTAAGGCTCATTGCTCAAAGGTATTTATTACAGTATCAGGA GCCAATACCTTGTGAGCAGTTGGTTACAGCGCTGTGTGATATCAAACAAGCTTATACACAATTTGGAGGAAAACGT CCCTTTGGTGTTCATTGCTGTACATTGGCTGGGATAAGCACTATGGCTTTCAGCTCTATCAGAGTGACCCTAGTGG AAATTACGGGGGATGGAAGGCCACATGCATTGGAATAATAGCGCTGCAGCTGTGTCAATGTTGAAACAAGACTAT AAAGAAGGAGAAATGACCTTGAAGTCAGCACTTGCTTTAGCTATCAAAGTACTAAATAAGACCATTGGATGTTAGTA AACTCTCTGCTGAAAAAGTGGAAATTGCAACACTAACAAGAGAGAATGGAAGACAGTAATCAGAGTTCTCAAA CAAAAAGAAGTGAGCAGTTGATCAAAAAACATGAGGAAGAAGAAGCCAAAGCTGAGCGTGAGAAGAAAAGAAA AAGAACAGAAAGAAAAGGATAAACGATCCGGAGGAGGAGGGTCCGGTGGGGGAGGCTCTGGAGGGGGTGGGTC TCGATCCATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCAGCGCGGGCCTCTGTGAAAACCTGTCTT GCACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCAGTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAAT GGATCGGCTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCAGGGCAAGGCCACCATGACCG CCGACACCAGCAGCAATACCGCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCC ACGACGCCATCACACGCCACCAGAACTTCGAGCTGTGGGGCGCCGCGCAACACCTGTGCTTCTTGGCG GCGGAGGCAGCGCGGAGGCGGAGCGGCGGCGGCGGAGCGGAGGCGGCTCTGATGTGGTGTGATGACCCAGAC ACCTCTGAGCCTGCCTGTGTCCCTGGGCGACAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAG CAACGGCAACACCTACCTGCAGTGGTACCTGCAAAAGCCCGGCCAGTCTCAAAGCTGCTGATGTACAAGGTGTC CAATGATTACGCGGTGTTCTGTATAGATTTAGCGGAGCGGAGCGGAACCGACTTACCCTGAAGATCAGCAG AGTGGAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTCACCTTCGGCTCCGGCAC CAAGCTCGAGATCAAG
PSMA4-	ATGTCTCGAAGATATGACTCCAGGACCACTATATTTTCTCCAGAAGGTGCGTTATACCAAGTTGAATATGCCATGGA AGCTATTGGACATGCAGGCACCTGTTTGGGAATTTTAGCAAATGATGGTGTGTTTGTCTGCAGCAGAGAGACGCAAC

(GGGGS)₆- scFv(c-Fos)	ATCCACAAGCTTCTTGATGAAGTCTTTTTTCTGAAAAAATTTATAAACTCAATGAGGACATGGCTTGCACTGTGGC AGGCATAACTTCTGATGCTAATGTTCTGACTAATGAACAAAGGCTCATTGCTCAAAGGTATTTATTACAGTATCAGGA GCCAATACCTTGTGAGCAGTTGGTTACAGCGCTGTGTGATATCAAACAAGCTTATACACAATTTGGAGGAAAAACGT CCCTTTGGTGTTCATTGCTGTACATTGGCTGGGATAAGCACTATGGCTTTCAGTCTTATCAGAGTGACCCCTAGTGG AAATTACGGGGGATGGAAGGCCACATGCATTGGAATAATAGCGCTGCAGCTGTGTCAATGTTGAAACAAGACTAT AAAGAAGGAGAAATGACCTTGAAGTCAGCACTTGCTTTAGCTATCAAAGTACTAAATAAGACCATTGGATGTAGTA AACTCTCTGCTGAAAAAGTGGAAATTGCAACACTAACAAGAGAGAATGGAAGACAGTAATCAGAGTTCTCAAA CAAAAAGAAGTGGAGCAGTTGATCAAAAAACATGAGGAAGAAGAAGCCAAAGCTGAGCGTGAGAAGAAAGAAA AAGAACAGAAAAGAAAGGATAAAAGATCCGGCGGAGGTGGCAGCGGTGGTGGCGGATCTGGCGGAGGTGGCTCT GGAGGAGGCGGTTCTGGCGGAGGCGGCTCAGTGGAGGTGGCAGCCGATCCATGGAAGTGCAGCTGCAGCAGAG CGGAGCCGAGCTGGTGGCGAGCGGGGCTCTGTGAAACTGTCTTGACCGCTTCTGGCTTCAACATCAAGGACTA CTACATGCACTGGGTCAAGCAGAGACCTGAGCAGGCGCTGGAATGGATCGGCTGGATGGACCCCAAGAACGGCG ATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCTGACACCGCCGACACCAAGCACTACCGCTTACCTGCG AGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCCATCACACGCCACCAGAACT TCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAGGCGAGCGGCGGAGGCGCAGCGGC GGCGGCGGCGAGCGGAGGCGGCTCTGATGTGGTGTGATGCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGAC CAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGTGACAGCAACGGCAACCTCAGCTTACCTGT CAAAAGCCCGGCCAGTCTCAAAGCTGCTGATGTACAAGGTGTCCAATAGATTACGCGGTGTTCTGTATAGATTTA GCGGACGCGGGAGCGGAACCGACTTCACCTGAAGATCAGCAGAGTGGAAGCCGAAGATCTGGGAGTGTACTTC TGCAGCCAGTCAACACACGTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATCAAG
PSMB2- scFv(c-Fos)	ATGGAGTACCTCATCGGTATCCAAGGCCCGACTATGTTCTTGTGCGCTCCGACCGGTGGCCGCCAGCAATATTGT CCAGATGAAGGACGATCATGACAAGATGTTTAAAGATGAGTGAAAAGATATTACTCCTGTGTGTTGGAGAGGCTGGA GACACTGTACAGTTTGCAGAATATATTAGAAAAACGTGCACTTTATAAGATGCGAAATGGATATGAATTGTCTCC CACGGCAGCAGCTAACTTCACACGCCGAAACCTGGCTGACTGTCTTCGGAGTCGGACCCCATATCATGTGAACCTC CTCCTGGCTGGCTATGATGAGCATGAAGGGCCAGCGCTGTATTACATGGACTACCTGGCAGCCTTGCCCAAGGCC CTTTTGCAGCCACGGCTATGGTGCCTTCTGACTCTCAGTATCCTCGACCGATACTACACACCGACTATCTACGT GAGAGGGCAGTGGAACCTCTTAGGAAATGTCTGGAGGAGCTCCAGAAACGCTTCATCTGAATCTGCCAACCTTC AGTGTTTCAATCATTGACAAAAATGGCATCCATGACCTGGATAACATTTCTTCCCAACAGGGCTCCAAAGCTTG GTACCGAGCTCGGATCCATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGGGAGCGGGGCTCTGTG AAACTGTCTTGACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCACTGGGTCAAGCAGAGACCTGAGCAG GGCTTGAATGGATCGGCTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCCTAAGTTCCAGGGCAAGGCC ACCATGACCGCCGACACCAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTG TACTACTGCCACGACGCCATCACACGCCACCAGAACTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTG TCTTCTGGCGGCGGAGGCGAGCGGCGGAGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGGCGG GACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGACAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCT GGTGACAGCAACGGCAACACCTACCTGCAGTGGTACCTGCAAAAGCCCGCCAGTCTCAAAGCTGCTGATGTA CAAGGTGTCCAATAGATTACGCGGTGTTCTGTATAGATTAGCGGCGAGCGGGAGCGGAACCGACTTCACCTTGAA GATCAGAGAGTGGAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTACCTTCGG CTCCGGCACCAAGCTCGAGATCAAG
PSMB2- (GGGGS)₃- scFv(c-Fos)	ATGGAGTACCTCATCGGTATCCAAGGCCCGACTATGTTCTTGTGCGCTCCGACCGGTGGCCGCCAGCAATATTGT CCAGATGAAGGACGATCATGACAAGATGTTTAAAGATGAGTGAAAAGATATTACTCCTGTGTGTTGGAGAGGCTGGA GACACTGTACAGTTTGCAGAATATATTAGAAAAACGTGCACTTTATAAGATGCGAAATGGATATGAATTGTCTCC CACGGCAGCAGCTAACTTCACACGCCGAAACCTGGCTGACTGTCTTCGGAGTCGGACCCCATATCATGTGAACCTC CTCCTGGCTGGCTATGATGAGCATGAAGGGCCAGCGCTGTATTACATGGACTACCTGGCAGCCTTGCCCAAGGCC CTTTTGCAGCCACGGCTATGGTGCCTTCTGACTCTCAGTATCCTCGACCGATACTACACACCGACTATCTACGT GAGAGGGCAGTGGAACCTCTTAGGAAATGTCTGGAGGAGCTCCAGAAACGCTTCATCTGAATCTGCCAACCTTC AGTGTTTCAATCATTGACAAAAATGGCATCCATGACCTGGATAACATTTCTTCCCAACAGGGCTCCAAAGCTTG GTACCGAGCTCGGATCCGGAGGAGGAGGTCCGGTGGGGGAGGCTCTGGAGGGGTGGGTCTGGATCCATGGAA GTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGGCGAGCGGGGCTCTGTGAAACTGTCTTGACCGCTTCTGG CTTCAACATCAAGGACTACTACATGCACTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGAT GGACCCCAAGAACGGCGATACCGAGTACGCCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACACCAGCA GCAATACCGCTTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCCATCA CCACGCCCACCAGAACTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAGGCGAGC GGCGGAGGCGGCGAGCGG GCCTGTGTCCCTGGGCGACAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGCAACA CTACCTGCAGTGGTACCTGCAAAAGCCCGGCCAGTCTCCAAAGCTGCTGATGTACAAGGTGTCCAATAGATTGAG CGGTGTTCTGTATAGATTAGCGGCGAGCGGGAGCGGAACCGACTTCACCTGAAGATCAGCAGAGTGGAAGCCGA AGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATC AAG
PSMB2- (GGGGS)₆- scFv(c-Fos)	ATGGAGTACCTCATCGGTATCCAAGGCCCGACTATGTTCTTGTGCGCTCCGACCGGTGGCCGCCAGCAATATTGT CCAGATGAAGGACGATCATGACAAGATGTTTAAAGATGAGTGAAAAGATATTACTCCTGTGTGTTGGAGAGGCTGGA GACACTGTACAGTTTGCAGAATATATTAGAAAAACGTGCACTTTATAAGATGCGAAATGGATATGAATTGTCTCC CACGGCAGCAGCTAACTTCACACGCCGAAACCTGGCTGACTGTCTTCGGAGTCGGACCCCATATCATGTGAACCTC CTCCTGGCTGGCTATGATGAGCATGAAGGGCCAGCGCTGTATTACATGGACTACCTGGCAGCCTTGCCCAAGGCC CTTTTGCAGCCACGGCTATGGTGCCTTCTGACTCTCAGTATCCTCGACCGATACTACACACCGACTATCTACGT GAGAGGGCAGTGGAACCTCTTAGGAAATGTCTGGAGGAGCTCCAGAAACGCTTCATCTGAATCTGCCAACCTTC AGTGTTTCAATCATTGACAAAAATGGCATCCATGACCTGGATAACATTTCTTCCCAACAGGGCTCCAAAGCTTG GTACCGAGCTCGGATCCGGAGGAGGAGGTCCGGTGGGGGAGGCTCTGGAGGGGTGGGTCTGGATCCATGGAA GTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGGCGAGCGGGGCTCTGTGAAACTGTCTTGACCGCTTCTGG CTTCAACATCAAGGACTACTACATGCACTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGAT GGACCCCAAGAACGGCGATACCGAGTACGCCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACACCAGCA GCAATACCGCTTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCCATCA CCACGCCCACCAGAACTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAGGCGAGC GGCGGAGGCGGCGAGCGG GCCTGTGTCCCTGGGCGACAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGCAACA CTACCTGCAGTGGTACCTGCAAAAGCCCGGCCAGTCTCCAAAGCTGCTGATGTACAAGGTGTCCAATAGATTGAG CGGTGTTCTGTATAGATTAGCGGCGAGCGGGAGCGGAACCGACTTCACCTGAAGATCAGCAGAGTGGAAGCCGA AGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATC AAG

	TCTGGCGGAGGCGGCTCAGGTGGAGGTGGCAGCGGATCCATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCGGAGCGGGGCCCTCTGTGAAACTGTCTTGCACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCACTGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACACAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCCATCACCACGCCCACCAGAAAACCTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAGGCGAGCGGCGGCGGCGGCGAGCGGAGGCGGCTCTGATGTGGTGATGACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGACCAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGCCAGTCTCCAAAGCTGTGTATGTACAAGGTGTCCAATAGATTACGCGGTGTTTCTGTATAGATTAGCGGCAGCGGGAGCGGAACCGACTTACCCTGAAGATCAGCAGAGTGGAAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATCAAG
scFv(c-Fos)- PSMC5	ATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCGGAGCGGGGCCCTCTGTGAAACTGTCTTGCACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCCTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACACAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCATCACCACGCCCACCAGAAAACCTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAGGCAGCGGCGGAGGCGGCGAGCGGCGGCGGCGGCGGCGGCGGCGGCGGCTGTATGTGGTGATGACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGACCAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGCAACACCTACCTGCAGTGGTACCTGCAAAAAGCCCGCCAGTCTCCAAAGCTGTGTATGTACAAGGTGTCCAATAGATTCAGCGGTGTTCTGTATAGATTAGCGGCAGCGGAGCGGGAACCGACTTACCCTGAAGATCAGCAGAGTGGAAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATCAAGGAATTCATGGCGCTTGACGGACCAGAGCAGATGGAGCTGGAGGAGGGGAAGGCAGGCAGCGGACTCGCCAATATTATCTGTCCAAGATTGAAGAATCCAGCTGATTGTGAATGATAAGAGCCAAAACCTCCGGAGGCTGCAGGCACAGAGGAACGAACATAAATGCTAAAGTTCGCCCTATTGCGGGAGGAGCTACAGCTGCTGCAGGAGCAGGGCTCCTATGTGGGGGAAGTAGTCCGGGCCATGGATAAGAAGAAAGTGTGGTCAAGGTACATCTGAAGGTAAATTTGTGTAGACGTGGACAAAAACATTGACATCAATGATGTGCACCCCAATTGCGGGTGGCTCTAAGGAATGACAGCTACTCTGCACAAGATCCTGCCCAACAAGGTAGACCCATTAGTGTCACTGATGATGGTGGAGAAAAGTACCAGATTCACTTATGAGATGATTGGTGGACTGGACAAACAGATCAAGGAGATCAAGAAGTGATCGAGCTGCCTGTAAAGCATCCTGAGCTCTTCAAGCACTGGGCATTGCTCAGCCCAAGGGAGTGTGTGTATGGACCTCCAGGCACTGGGAAGACACTGTTGGCCCGGGCTGTGGCTCATCATACGGACTGTACCTTTATTCGTGTCTCTGGCTCTGAACCTGGTACAGAAATTCATAGGGGAAGGGGCAAGAATGGTGAGGGAGCTGTTTGTCTATGGCACGGGAACATGCTCCATCTATCATCTTCATGGACGAAATCGACTCCATCGGCTCCTCGCGGCTGGAGGGGGTCTGGAGGGGACAGTGAAGTGCAGCGCACGATGCTGGAGTTGCTCAACCAGCTCGACGGCTTTGAGGCCACCAAGAACATCAAGGTTATCATGGCTACTAATAGGATTGATATCTGGACTCGGCCTGCTTCCGCGAGCGGGAACCGACTTCAAGGAAATTTGAATGCCACCCCGGAGGAGGCCCGGCTGGACATTTTGAAGATTCATTCTCGGAAGATGAACCTGACCCGGGGGATCAACCTGAGAAAAATTGCTGAGCTCATGCCAGGAGCATAGGGGCTGAAGTGAAGGCGGTGTGCACAGAAGCTGGCATGTATGCCCTGCGAGAACGGCGAGTCCATGTCACTCAGGAGGACTTTGAGATGGCAGTAGCCAAGGTCATGCAGAAGGACAGTGAGAAAACATGTCCATCAAGAAATTATGGAAG
scFv(c-Fos)- (GGGS) ₃ - PSMC5	ATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCGGAGCGGGGCCCTCTGTGAAACTGTCTTGCACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCCTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACACAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCATCACACGCGCCACCAGAAAACCTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAGGCAGCGGCGGAGGCGGCGAGCGGCGGCGGCGGCGGCGGAGGCGGCTCTGTATGTGGTGATGACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGACCAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGCAACACCTACCTGCAGTGGTACCTGCAAAAAGCCCGCCAGTCTCCAAAGCTGTGTATGTACAAGGTGTCCAATAGATTCAGCGGTGTTCTGTATAGATTAGCGGCAGCGGGAACCGACTTACCCTGAAGATCAGCAGAGTGGAAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATCAAGGAATTCGGAGGAGGAGGGTCCGGTGGGGGAGGCTCTGGAGGGGGTGGGTCTGAATTCATGGCGCTTGACGGACCAGAGCAGATGGAGCTGGAGGAGGGGAAGGCAGGCAGCGGACTCCGCCAATATTATCTGTCCAAGATTGAAGAACTCCAGCTGATTGTGAATGATAAGAGCCAAAACCTCCGAGGCTGCAGGCACAGAGGAACGAACATAATGCTAAAGTTGCCTATTGCGGGAGGAGCTACAGCTGTGTCAGGAGCAGGGCTCCTATGTGGGGGAAGTAGTCCGGCCATGGATAAGAAGAAAGTGTGGTCAAGGTACATCCTGAAGGTAAATTTGTTGTAGACGTGGACAAAAACATGACATCAATGATGTGACACCAATTGCCGGGTGGCTCTAAGGAATGACAGCTACACTCTGCACAAGATCCTGCCCAACAAGGTAGACCCATTAGTGTCACTGATGATGGTGGAGAAAGTACCAGATTCACTTATGAGATGATTGGTGGACTGGACAAAACAGATCAAGGAGATCAAGAAGTGATCGAGCTGCCGTGTTAAGCATCCTTGAGCTCTTCGAAAGCACTGGCATTGCTCAGCCCAAGGGAGTGTGTGTATGGACCTCCAGGCACTGGGAAGACACTGTTGGCCCGGGCTGTGGCTCATCATACGGACTGTACCTTTATTCGTGTCTCTGGCTCTGAACCTGGTACAGAAATTCATAGGGGAAGGGGCAAATGATGGTGAAGGGAGCTGTTTGTATGGCACGGGAACATGCTCCATCTATCATCTTCATGGACGAAATCGACTCCATCGGCTCCTCGCGGCTGGAGGGGGTCTGGAGGGGACAGTGAAGTGCAGCGCACGATGCTGGAGTTGCTCAACCAGCTCGACGGCTTTGAGGCCACCAAGAACATCAAGGTTATCATGGCTACTAATAGGATTGATATCTGGACTCGGCCTGCTTCCGCGAGCGGGAACCGACTTCAAGGAAATTTGAATGCCACCCCGGAGGAGGCCCGGCTGGACATTTTGAAGATTCATTCTCGGAAGATGAACCTGACCCGGGGGATCAACCTGAGAAAAATTGCTGAGCTCATGCCAGGAGCATAGGGGCTGAAGTGAAGGCGGTGTGCACAGAAGCTGGCATGTATGCCCTGCGAGAACGGCGAGTCCATGTCACTCAGGAGGACTTTGAGATGGCAGTAGCCAAGGTCATGCAGAAGGACAGTGAGAAAACATGTCCATCAAGAAATTATGGAAG
scFv(c-Fos)-	ATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCGGAGCGGGGCCCTCTGTGAAACTGTCTTGCACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCCTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGGCTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACACAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGCATCACACGCGCCACCAGAAAACCTTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAGGCAGCGGCGGAGGCGGCGAGCGGCGGCGGCGGCGGCGGAGGCGGCTCTGTATGTGGTGATGACCCAGACACCTCTGAGCCTGCCTGTGTCCCTGGGCGACCAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGCAACACCTACCTGCAGTGGTACCTGCAAAAAGCCCGCCAGTCTCCAAAGCTGTGTATGTACAAGGTGTCCAATAGATTCAGCGGTGTTCTGTATAGATTAGCGGCAGCGGGAACCGACTTACCCTGAAGATCAGCAGAGTGGAAAGCCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCTTTACCTTCGGCTCCGGCACCAAGCTCGAGATCAAGGAATTCGGAGGAGGAGGGTCCGGTGGGGGAGGCTCTGGAGGGGGTGGGTCTGAATTCATGGCGCTTGACGGACCAGAGCAGATGGAGCTGGAGGAGGGGAAGGCAGGCAGCGGACTCCGCCAATATTATCTGTCCAAGATTGAAGAACTCCAGCTGATTGTGAATGATAAGAGCCAAAACCTCCGAGGCTGCAGGCACAGAGGAACGAACATAATGCTAAAGTTGCCTATTGCGGGAGGAGCTACAGCTGTGTCAGGAGCAGGGCTCCTATGTGGGGGAAGTAGTCCGGCCATGGATAAGAAGAAAGTGTGGTCAAGGTACATCCTGAAGGTAAATTTGTTGTAGACGTGGACAAAAACATGACATCAATGATGTGACACCAATTGCCGGGTGGCTCTAAGGAATGACAGCTACACTCTGCACAAGATCCTGCCCAACAAGGTAGACCCATTAGTGTCACTGATGATGGTGGAGAAAGTACCAGATTCACTTATGAGATGATTGGTGGACTGGACAAAACAGATCAAGGAGATCAAGAAGTGATCGAGCTGCCGTGTTAAGCATCCTTGAGCTCTTCGAAAGCACTGGCATTGCTCAGCCCAAGGGAGTGTGTGTATGGACCTCCAGGCACTGGGAAGACACTGTTGGCCCGGGCTGTGGCTCATCATACGGACTGTACCTTTATTCGTGTCTCTGGCTCTGAACCTGGTACAGAAATTCATAGGGGAAGGGGCAAATGATGGTGAAGGGAGCTGTTTGTATGGCACGGGAACATGCTCCATCTATCATCTTCATGGACGAAATCGACTCCATCGGCTCCTCGCGGCTGGAGGGGGTCTGGAGGGGACAGTGAAGTGCAGCGCACGATGCTGGAGTTGCTCAACCAGCTCGACGGCTTTGAGGCCACCAAGAACATCAAGGTTATCATGGCTACTAATAGGATTGATATCTGGACTCGGCACTGCTTCGCCCAGGGCGCATGACAGAAAAATTGAATTCCACCCCCCAATGAGGAGGCCGGCTGGACATTTGAAGATTCATTCTCGGAAGATGAACCTGACCCGGGGGATCAACCTGAGAAAAATTGCTGAGCTCATGCCAGGAGCATAGGGGCTGAAGTGAAGGCGGTGTGCACAGAAGCTGGCATGTATGCCCTGCGAGAACGGCGAGTCCATGTCACTCAGGAGGACTTTGAGATGGCAGTAGCCAAGGTCATGCAGAAGGACAGTGAGAAAACATGTCCATCAAGAAATTATGGAAG
scFv(c-Fos)-	ATGGAAGTGCAGCTGCAGCAGAGCGGAGCCGAGCTGGTGCGGAGCGGGGCCCTCTGTGAAACTGTCTTGCACCGCTTCTGGCTTCAACATCAAGGACTACTACATGCCTGGGTCAAGCAGAGACCTGAGCAGGGCCTGGAATGGATCGG

(GGGS)₆- PSMC5	CTGGATGGACCCCAAGAACGGCGATACCGAGTACGCCCTAAGTTCCAGGGCAAGGCCACCATGACCGCCGACAC CAGCAGCAATACCGCCTACCTGCAGCTGAGCAGCCTTACAAGCGACGACACAGCCGTGTACTACTGCCACGACGC CATCACCACGCCCACCAGAAATTCGACGTGTGGGGCGCCGGCACAACCGTGACCGTGTCTTCTGGCGGCGGAG GCAGCGGCGGAGGCGGCAGCGGCGGCGGCGGAGCGGAGCGCGGCTCTGATGTGGTGATGACCCAGACACCTCTG AGCCTGCCTGTGTCCCTGGGCGACAGGCCAGCATCAGCTGTAGAAGCTCTCAGAGCCTGGTGCACAGCAACGGC AACACCTACCTGCACTGGTACCTGCAAAAAGCCCGCCAGTCTCCAAAGCTGTGTATGTACAAGGTGTCCAATAGAT TCAGCGGTGTTCTGTATAGATTTAGCGGCAGCGGGAGCGGAACCGACTTCACCCTGAAGATCAGCAGAGTGGAAG CCGAAGATCTGGGAGTGTACTTCTGCAGCCAGTCAACACACGTGCCCTTTCACCTTCGGCTCCGGCACCAAGCTCG AGATCAAGGAATTTCGGCGGAGGTGGCAGCGGTGGTGGCGGATCTGGCGGAGGTGGCTCTGGAGGAGGCGGTTCT GGCGGAGGCGGCTCAGGTGGAGGTGGCAGCGAATTCATGGCGCTTGACGGACCAGAGCAGATGGAGCTGGAGGA GGGGAAGGCAGGCAGCGGACTCCGCCAATATTATCTGTCCAAGATTGAAGAACTCCAGCTGATTGTGAATGATAAG AGCCAAAACCTCCGGAGGCTGCAGGCACAGAGGAACGAATAATGCTAAAGTTTCGCTATTGCGGGAGGAGCT ACAGCTGCTGCAGGACGAGGCTCCTATGTGGGGAAAGTAGTCCGGGCCATGGATAAGAAGAAAGTGTGGTGCA GGTACATCCTGAAGGTAAATTTGTTGTAGACGTGGACAAAAACATTGACATCAATGATGTGACACCCAATTGCCGG GTGGCTCTAAGGAATGACAGCTACACTCTGCACAAGATCTGCCCAACAAGGTAGACCCATTAGTGTCACTGATGA TGGTGGAGAAAGTACCAGATTCAACTTATGAGATGATTGGTGGACTGGACAAACAGATCAAGGAGATCAAAGAAG TGATCGAGCTGCCTGTTAAGCATCCTGAGCTTCTCGAAGGACCTGGGCAATGTCTGACCCAAAGGAAAGTGTGTA TGGACCTCCAGGCACTGGGAAGACACTGTTGGCCCGGCTGTGGCTCATCATACGCACTGTACCTTTATTCGTGTC TCTGGCTCTGAACTGGTACAGAAATTCATAGGGGAAGGGGCAAGAATGGTGAGGGAGCTGTTTGTCTATGGCACGG GAACATGCTCCATCTATCATCTTCATGGACGAAATCGACTCCATCGGCTCCTCGCGCTGGAGGGGGTCTGGAG GGGACATGAAAGTGCAGCGCACGATGCTGGAGTTGCTCAACAGCTCGACCGCTTGAAGGCAACGAAGAACTC AAGGTTATCATGGCTACTAATAGGATTGATATCCTGGACTCGGCAGTCTTCGCCAGGGCGCATTGACAGAAAAAT TGAAATCCCAACCCCAATGAGGAGGCGCGGCTGGACATTTGAAGATTATTCTCGGAAGATGAACCTGACCCGG GGGATCAACCTGAGAAAAATTGCTGAGCTCATGCCAGGAGCATCAGGGGCTGAAGTGAAGGCGGTGTGCACAGA AGCTGGCATGTATGCCCTGCGAAGCGGAGTCTACTCAGGAGGACTTTGAGATGGCAGTAGCCAAGGT CATGCAGAAGGACAGTGAGAAAAACATGTCCATCAAGAAATTATGGAAG
PSMD4- scFv(Flag)	ATGGTGTGGAAAGCACTATGGTGTGTGGACAACAGTGAGTATATGCGGAATGGAGACTTCTTACCCACCAGGC TGCAGGCCCAGCAGGATGCTGTCAACATAGTTGTCTATTCAAAGACCCGCAGCAACCTGAGAACAACGTGGGCC TTATCACACTGGCTAATGACTGTGAAGTGCTGACCACACTACCCAGACACTGGCCGTATCCTGTCCAAGCTACA TACTGTCCAACCCCAAGGGCAAGATCACCTTCTGCAGGCAATCCGCGTGGCCCATCTGGCTCTGAAGCACCCGACA AGGCAAGAATCACAAGATGCGCATCATTGCCCTTTGTGGGAAGCCAGTGGAGGACAATGAGAAGGATCTGGTGAA ACTGGCTAAACGCCTCAAGAAGGAGAAAAGTAAATGTTGACATTATCAATTTGGGGAAGAGGAGGTGAACACAGA AAAGCTGACAGCCTTTGTAAACACGTTGAATGGCAAGATGGAACCGGTTCTCATCTGGTGACAGTGCCTCTGG GCCAGTTTGGCTGATGCTCTCATCAGTTCTCCGATTTTGGCTGGTGAGGTGGTGCCATGCTGAGTCTTGGTGCC AGTGACTTTGAATTTGGAGTAGATCCAGTGCTGATCCTGAGCTGGCCTTGGCCCTTCGTGTATCTATGGAAGAGC AGCGCAGCGGCAGGAGGAGGAGGCGCGGCGGCGAGCTGCAGCTTCTGCTGCTGAGGCCGGGATTGTACGACT GGGACTGAAGACTCAGACGATGCCCTGCTGAAGATGACCATCAGCCAGCAAGAGATTGGCCGCACTGGGCTTCCT GACCTAAGCAGTATGACTGAGGAAGAGCAGATTGCTTATGCCATGCAGATGTCCCTGACGGGAGCAGAGTTTGGC CAGGCGGAATCAGCAGACATTGATGCCAGCTCAGCTATGGACACATCTGAGCCAGCCAGGAGGAGGATGTATTAC GACGTGATGCAGGACCCGAGTTCCTTCAGAGTGTCTAGAGAACCTCCCAGGTGTGGATCCCAACAATGAAGCC ATTCGAAATGCTATGGGTCCCTGGCTCCCAGGCCACCAAGGACGGCAAGAAGGACAAGAAGGAGGAAGACAA GAAGCGACCCCTTAATTAACGAGGTGAAGCTGGTGGAGTCTGGGGGAGGCTTAGTGAAGCCTGGAGGGTCCCTGA AACTCTCCTGTGCAGCCTCTGGATTCACTTTCAGTAGTTTGGAAATGCAGTGGGTTCCGCACTCCGAGCTCCGAGAG GCTGGAGTGGGTGCGATACATAAGTAGTGGCAGTAGTACCATCTACTATGCAGACACAGTGAAGGGCCGATTACCC ATCTCCAGAGACAATGCCAAGAACACCTGTACCTGCAAATGAGCAGTCTGAGGTCTGAGGACACGGCCATATATT ACTGTGCAAGATCCCTTGCTACGGCCGCTTTTGCTTACTGGGGCCAAGGCACCACCTCAGAGTCTCCTCAGGTG AGCGGTTTCAAGGCGGAGGTGGCTCTGGCGGTGGCGGATCGGCATCGTTCTGACACAGTCCCCGCGACCTTAC TGCTCACTCGGCCAGAGGGCTACCATTTCTGCAGATCCTCTCAAAGCATAGTGTACAGTAATGAAAAACCTATT TAGAATGGTACCAACAGAAACCAGGACAGCCACCCAACTCCTCATCTATAAAGTTTCCAACCGATTTTCTGGGAT CCCAGCCAGGTTTATGTGGCAGTGGGTCTGGGACAGCACTTACCCTCAACATCCATCCTGTGGAGGAGGAGGATGC TGCAACCTATTACTGTTTCAAGGTTACATGTTCCGTCACGTTTCGGTGCTGGGACCAAGCTGGAAATAAAA
scFv(HA)- ADRM1	ATGGAGGTGAAGCTGGTGGAGTCTGGGGGAGGCTTAGTGAAGCCTGGAGGGTCCCTGAAACTCTCCTGTGCAGC CTCTGGATTCACTTTTCAGTAGTTATGGCATGTCTTGGGTTCCGACAGCTCCGGAGAAGAGGCTGGAGTGGGTGCGA ACCATAGTAGAGGTGGTAGTTATACCTACTATCCAGACAGTGTGAAGGGGCGATTCAACATCTCCAGAGACAATG CCAAGAACACCCGTGACCTGCAATGAGCAGTCTGAGGTCTGAGGACACGGCCATATATTACTGTGCAAGACGGG AAACATACGATGAAAAGGGCTTTGCCCTACTGGGGCCAAGGCACCCTCTCACAGTCTCCTCAGGTGGAGGCGGTT CAGGCGGAGGTGGCTCTGGCGGTGGCGGATCGGACATTGTGCTGACCCAATCTCCAGCTTCTTTGACTGTGTCTCT AGGGCAGAGGGCCACCATCTCCTGCAAGTCCAGCCAAAGCTTCTGAATAGTGGTAATCAGAAGAACTACCTCAC CTGGTACCAACAGAAACCAGGACAGCCACCCAACTCCTCATCTATTGGGCATCCACTAGGGAATCTGGGATCCCCA GCCAGGTTTAGTGGCAGTGGGTCTGGGACAGACTTACCCTCAACATCCATCCTGTGGAGGAGGAGGATGCTGCA ACCTATTACTGTGAGAACGATAATTCCCATCCGCTCACGTTCCGGTGTGGGACCAAGCTGGAATAAAAAAGCGCGC CAATGACGACCTCAGGCGCGCTCTTCCAAGCCTGGTGCCAGGCTCTCGGGGCGCTCCAAACAAGTACTTGGTGG AGTTTCGGGCGGGAAAGATGTCCCTGAAGGGGACCACCGTGACTCCGGATAAGCGGAAAGGGCTGGTGTACATTC AGCAGCGGACGACTCGTTATTCACTTCTGCTGGAAGGACAGGACGTCCGGGAACGTGGAAGCAGGATGTCA TCTTCCCTGACGACTGTGAGTTCAAGCGGGTGGCCGAGTGCCCCAGCGGGAGGGTCTACGTGTGAAGTTCAAGG CAGGTTCCAAGCGGCTTTTCTTCTGGATGCAGGAACCCAAGACAGACCAGGATGAGGAGCATTGCCGGAAGTC AACGAGTATCTGAACAACCCCCGATGCCTGGGGCGCTGGGGGCCAGCGGAAGCAGCGGCCACGAACTCTCTGC GCTAGGCGGTGAGGGTGGCCTGCAGAGCCTGCTGGGAAACATGAGCCACAGCCAGCTCATGCAGCTCATCGGACC

	AGCCGGCCTTGGAGGACTGGGTGGGCTGGGGGCCCTGACTGGACCTGGCCTGGCCAGCTTACTGGGGAGCAGTG GGCCTCCAGGGAGCAGCTCCTCCTCCAGCTCCCGAGCCAGTCGGCAGCGGTACCCCCGTCATCCACCACCTCTT CCACCCGTGCCACCCAGCCCCCTTCTGCTCCAGCAGCTGCCCTAGCAACTAGCCCCAGCCCCGCGCCAGTTCCG GGAATGGAGCCAGCACAGCAGCCAGCCCCAGCCCATCCAGCTGAGCGACCTCCAGAGCATCTGAGCCACG ATGAACGTACCAGCCGGGCCAGCAGGCGGCCAGCAAGTGGACCTGGCCAGTGTGCTGACGCCGGAGATAATGGC TCCCATCCTCGCCAACGCGGATGTCCAGGAGCGCTGCTTCCCTACTTGCCATCTGGGGAGTCGCTGCCGAGACC GCGGATGAGATCCAGAATACCTGACCTCGCCCCAGTTCCAGCAGGCCCTGGGCATGTTACGCGCAGCCTTGGCCT CGGGGAGCTGGGCCCCCTCATGTGCCAGTTCCGTCTGCCTGCAGAGGCTGTGGAGGCCGCCAACAAAGGGCGATG TGGAAGCGTTTGCCAAAGCCATGCAGAAACGCCAAGCCCCGAGCAGAAAGAGGGCGACACGAAGGACAAGAA GGACGAAGAGGAGGACATGAGCCTGGAC
VHH(GFP)- ADRM1	ATGGTCCAACCTGGTGGAGTCTGGTGGCGCTTTGGTGCAGCCAGGTGGCTCTCTGCGTTTTGCTGTGCCGCTTCTG GCTTCCAGTGAACCGCTATTCCATCGCTGGTATCGCCAGGCTCCAGGCAAAAGAGCGTGAGTGGGTAGCCGGTAT GTCCAGCGCGGGTGATCGTAGCTCCTATGAAGACTCCGTGAAGGGCCGTTTACCATCGACCGCTGACGATGCCGT AACACGGTGTATCTGCAAATGAACAGCTTGAACCTGAAGATACGGCCGTGTATTACTGTAATGTGAACGTGGGCT TCGAGTATTGGGGCCAAGGCACCCAGGTCACCGTCTCCAGCGGCGGCCAATGACGACCTCAGGCGCGCTCTTTC CAAGCCTGGTGCCAGGCTCTCGGGGCGCCTCCAACAAGTACTTGGTGGAGTTTCGGGCGGGAAAGATGTCCTGA AGGGGACCAACCGTACTCCGATAAGCGGAAAGGGCTGGTGATACATTACGACAGCAGCCGTGACCTTATCTACT TCTGCTGGAAGGACAGGACGTCCGGGAACGTGGAAGACGACTTGATCATCTTCCCTGACGACTGTGAGTTCAAGC GGTGCCCGCAGTGCCCCAGCGGGAGGGTCTACGTGCTGAAGTTCAAGGCAGGGTCCAAGCGGCTTTTCTTCTGGA TGCAGGAACCCAAGACAGACCAGGATGAGGAGCATTGCGGAAAGTCAACGAGTATCTGAACAACCCCCGATG CCTGGGGCGCTGGGGGCCAGCGGAAGCAGCGGCCAGCAACTCTGCGCTAGGCGGTGAGCTGATGCTTATCAGAG CCTGCTGGGAAACATGAGCCACAGCCAGCTCATGCAGCTCATCGGACCAGCGGCCCTTGAGGAGTGGGTGGGCT GGGGGCCCTGACTGGACCTGGCCTGGCCAGCTTACTGGGGAGCAGTGGGCTCCAGGGAGCAGCTCCTCCTCCA GCTCCCGAGCCAGTCGGCAGCGGTACCCCGTCACTCCACCACCTCTTCCACCCGTGCCACCCAGCCCCCTTCTGC TCCAGCAGCTGCCTCAGCAACTAGCCCCAGCCCCGCGCCAGTTCCGGGAATGGAGCCAGCACAGCAGCCAGCC CGACCCAGCCCATCCAGCTGAGCGACCTCAGAGCATCTTGGCCACGATGAACGTACCAGCCGGGCCAGCAGGCG GCCAGCAAGTGGACCTGGCCAGTGTGCTGACGCCGGAGATAATGGCTCCCATCCTCGCCAACGCGGATGTCCAGG AGCGCTGCTTCCCTACTTGCCATCTGGGGAGTCGCTGCCGAGACCGCGGATGAGATCCAGAATACCTGACCTC GCCCCAGTTCCAGCAGGCCCTGGGCATGTTACGCGCAGCCTTGGCCTCGGGGAGCTGGGCCCCCTCATGTGCCA GTTCCGGTCTGCCTGCAGAGGCTGTGGAGGCCGCCAACAAAGGGCGATGTGGAAGCGTTTGCCAAAGCCATGCAGA ACAACGCCAAGCCCCAGCAGAAAGAGGGCGACACGAAGGACAAGAAGGACGAAGAGGAGGACATGAGCCTGG AC
PSMD4- VHH(GFP)	ATGGTGTTGGAAAGCACTATGGTGTGTGTGGACAACAGTGAGTATATGCGGAATGGAGACTTCTTACCCACCAGGC TGACGGCCAGCAGGATGCTGTCAACATAGTTGTCAATCAAAGACCCGACGAACCCCTGAGAACAACGTGGGCC TTATCACACTGGCTAATGACTGTGAAGTGTGACCACACTCACCCAGACACTGGCCGTATCCTGTCCAAGCTACA TACTGTCCAACCCAAGGGCAAGATCACCTTCTGCACGGGCATCCGCGTGGCCCATCTGGCTCTGAAGCACCGACA AGGCAAGAATCACAGATGCGCATATTGCCTTTGTGGGAAGCCAGTGGAGGACATGAGAAGGATCTGGTGAA ACTGGCTAAACGCCTCAAGAAGGAGAAAAGTAAATGTTGACATTATCAATTTTGGGGAAGAGGAGGTGAACACAGA AAAGCTGACAGCCTTTGTAACACGTTGAATGGCAAGAGATGGAACCGGTTCTCATCTGGTGACAGTGCCCTCTGG GCCCAGTTTGGCTGATGCTCTCATCAGTTCTCCGATTTTGGCTGGTGAAGGTGGTGCCATGCTGGGTCTTGGTGCC AGTGACTTTGAATTTGGAGTAGATCCAGTGCTGATCTGAGCTGGCCTTGCCCTTCGTGTATCTATGGAAGAGC AGCGGCAGCGGCAGGAGGAGGAGGCCCGGGCAGCTGCAGCTTCTGCTGCTGAGGCCGGGATTGTCTACGACT GGGACTGAAGACTCAGACGATGCCCTGCTGAAGATGACCATGAGCCAGCAAGAGTTTGGCCGCACTGGGCTTCT GACCTAAGCAGTATGACTGAGGAAGAGCAGATTGCTTATGCCATGCAGATGTCCCTGCAGGGAGCAGAGTTTGGC CAGGCGGAATCAGCAGACATTGATGCCAGCTCAGCTATGGACACATCTGAGCCAGCCAAGGAGGAGGATGATTAC GACGTGATGCAGGACCCGAGTTCTTCAAGTGTCTAGAGAACCTCCCAGGTGTGGATCCCAACAATGAAGCC ATTGAAATGCTATGGGTCCCTGGCCCTCCAGGCCAAGGACGGCAAGAAGGACAAGGAGGAAGACAAGAA GAAGCGACCCTTAATTAACGTCCAACCTGGTGGAGTCTGGTGGCGCTTTGGTGCAGCCAGGTGGCTCTCTGCGTTTG TCCTGTGCCGCTTCTGGCTTCCAGTGAACCGCTATTCCATGCGCTGGTATGCCAGGCTCCAGGCAAAAGAGCGTG AGTGGGTAGCCGGTATGTCCAGCGCGGGTATCGTAGCTCCTATGAAGACTCCGTGAAGGGCCGTTTACCATCAG CCGTGACGATGCCGTAACACGGTGTATCTGCAAATGAACAGCTTGAACCTGAAGATACGGCCGTGTATTACTGT AATGTGAACGTGGGCTTCGAGTATTGGGGCCAAGCCAGCCAGGTCACCGTCTCCAG
ADRM1- scFv(BRD4)	ATGACGACCTCAGGCGCGCTCTTTCCAAGCCTGGTGCCAGGCTCTCGGGGCGCCTCCAACAAGTACTTGGTGGAG TTTCGGGCGGGAAAGATGTCCCTGAAGGGGACCACCGTGACTCCGGATAAGCGGAAAGGGCTGGTGTACATTCAG CAGACGGGAGCTCGCTTATCACTTCTGCTGGAAGGACAGGACGTCCGGGAACGTGGAAGACGACTTGATCATC TTCCCTGACGACTGTGAGTTCAAGCGGGTGCCGAGTGCCCCAGCGGGAGGGTCTACGTGCTGAAGTTCAAGGCA GGGTCCAAGCGGCTTTTCTTCTGGATGCAGGAACCCAAGACAGACCAGGATGAGGAGCATTGCCGGAAGTCAA CGAGTATCTGAACAACCCCCGATGCCTGGGGCGCTGGGGGCCAGCGGAAGCAGCGGCCACGAACCTCTGCGCT AGGCGGTGAGGGTGGCTGCAGAGCCTGCTGGGAAACATGAGCCACAGCCAGCTCATGCAGCTCATCGGACCAG CCGCTTGGAGGACTGGGTGGGCTGGGGGCCCTGAGTGACCTGGCCTGGCCGAGCTTACTGGGAGCAGTGGG CCTCCAGGGAGCAGCTCCTCCTCCAGCTCCCGGAGCCAGTCGGCAGCGGTACCCCGTCATCCACCACCTCTTCC ACCGTGCCACCCAGCCCCCTTCTGCTCCAGCAGCTGCCTCAGCAACTAGCCGAGCCCCGCGCCAGTTCGGG AATGGAGCCAGCACAGCAGCCAGCCGACCCAGCCATCCAGCTGAGCGACCTCCAGAGCATCCTGGCCACGATG AACGTACCAGCCGGGCCAGCAGGCGGCCAGCAAGTGGACCTGGCCAGTGTGCTGACGCCGGAGATAATGGCTCC CATCTCGCCAACGCGGATGTCCAGGAGCGCCTGATTCCTTACTTGGCATCTGGGGAGTCGCTGCCGAGCAGCGG GATGAGATCCAGAATACCTGACCTCGCCCCAGTTCCAGCAGGCCCTGGGCATGTTACGCGCAGCCTTGGCCTCGG GGCAGCTGGGCCCCCTCATGTGCCAGTTCGGTCTGCCTGCAGAGGCTGTGGAGGCCGCCAACAAAGGGCGATGTGG AAGCGTTTGCCAAAGCCATGCAGAAACGCCAAGCCCCGAGCAGAAAGAGGGCGACACGAAGGACAAGAAGGA

	CGAAGAGGAGGACATGAGCCTGGACTTAATTAACGAGGTGCAATTGTTGGAGAGCGGGGGAGGCTTGGTACAGC CTGGGGGGTCCCTGCGCCTCTCTGTGCAGCCAGCGGATTACCTTTTACGGTTATGGTATGTCTTGGGTCCGCCAG GCTCCAGGGAAGGGGCTGGAGTGGGTCTCATACATTGGTGGTTACGGTAGTTCTACATCTTATGCAGACTCCGTGA AGGGCCGGTTCACCATCTCCCGTGACAATTCCAAGAACACGCTGTATCTGCAAATGAACAGCCTGCGTGCCGAGG ACACGGCTGTATATTATTGTGCGCGCGGTTCTGCTTACATTGACTATTGGGGCCAAGGAACCCCTGGTCACCGTCTCC TCAGGTGGAGGCGGTTTCAGGCGGAGGTGGATCCGGCGGTGGCGGATCGGACATCCAGATGACCCAGTCTCCATCC TCCCTGAGCGCATCTGTAGGAGACCGCGTCACCATCACCTGCAGGGCAAGTCAGAGCATTAGCAGCTATTTAAATT GGTATCAGCAGAAACCAGGGAAGCCCCCTAAGCTCCTGATCTATGCTGCATCCAGTTTGCAAAGTGGGGTCCCATC ACGTTTCAGTGGCAGTGGAAGCGGGACAGATTTCACTCTCACCATCAGCAGTCTGCAACCTGAAGATTTTGC AAC TTATTACTGTCAACAGTACAACCTCTCTGTACACTTTTGGCCAGGGGACCAAGCTGGAGATCAAA
mutPSMD4(6)	ATGGTGTTGAAAGCACTATGGTGTGTGTGGCTAACGCTGCTGCTATGCGGAATGGAGACTTCTTACCCACCAGGC TGCAGGCCCAGCAGGATGCTGTCAACATAGTTTGTCAATCAAAGACCCGCAGCAACCCCTGAGAACAACTGGGGC TTATCACACTGGCTAATGACTGTGAAGTGTGACACACTCACCCAGACACTGGCCGTATCCTGTCCAAGCTTACA TACTGTCCAACCCAAGGGCAAGATCACCTTCTGCACGGGCATCCGCGTGGCCCATCTGGCTCTGAAGCACCGACA AGGCAAGAATCACAAGATGCGCATCATTTGCCCTTGTGGCTGCTCCAGTGGAGGACAATGAGAAGGATCTGGTGAA ACTGGCTAAACGCCTCAAGAAGGAGAAAAGTAAATGTTGACATTATCAATTTTGGGGAAGAGGAGGTGAACACAGA AAAGCTGACAGCCTTTGTAACACGTTGAATGGCAAAAGATGGAACCGGTTCTCATCTGGTGACAGTGCCTCCTGG GCCCAGTTTGGCTGATGCTCTCATCAGTTCTCCGATTTTGGCTGGTGAAGGTGGTGCCATGCTGGGTCTTGGTGCC AGTGACTTTGAATTTGGAGTAGATCCAGTGCTGATCCTGAGCTGGCCTTGGCCCTTCGTGTATCTATGGAAGAGC AGCGGCAGCGGCAGGAGGAGGAGGCCCGGCGGCGAGCTGCAGCTTCTGCTGCTGAGGCCGGGATTGCTACGACT GGGACTGAAGACTCAGACGATGCCCTGCTGAAGATGACCATAGCCAGCAAGAGATTTGGCCGCTAGCTGGGCTTCT GACCTAAGCAGTATGACTGAGGAAGAGCAGATTGCTTATGCCATGCAGATGTCCCTGCAGGGAGCAGAGTTTGGC CAGGCGGAATCAGCAGACATTGATGCCAGCTCAGCTATGGACACATCTGAGCCAGCCAAGGAGGAGGATGATTAC GACGTGATGCAGGACCCCGAGTTCTTTCAGAGTGTCTAGAGAACCCTCCAGGTGTGGATCCCAACAATGAAGCC ATTCGAAATGCTATGGGCTCCCTGGCCTCCCAGGCCACCAAGGACGGCAAGAAGGACAAGAAGGAGGAAGACAA GAAGCGACCC
ADRM1 (APru)	ATGCCGCTATGCCCGGAGCACTCGGAGCATCTGGGAGTAGTGGGCATGAGTTGTCCGCTCTCGGTGGAGAAGGC GGGTACAATCTCTTCTCGGGAATATGTCTCATTTCTCAACTGATGCAATTAATTGGGCCGGCTGGACTGGGCGGTCT CGGCGGCTTGGAGCATTGACCGGGCCAGGGCTCGTTTCACTCTTAGGATCTTCTGGTCCCCCGGGTAGTTCATCT TCATCATCAAGCCGATCACAAGTGTCTGTGTGACACCCCTTAGCACTACGAGCAGCACTAGGGCAACGCCTGCA CCTTCCGCACCTGCGGCAGCATCCGCCACATCACCTAGTCTGCTCCTAGCAGTGGCAACGGCGCAAGTACTGCCG CTTCTCCTACTCAACCAATTCAATTGAGTGATCTGCAAAAGTATTTGGCAACCATGAATGTCCCGCTGGACCTGCG GGCGGACAACAGGTGATCTCGCGAGCGTATTGACCCCCGAAATCATGGCCCCAATATTAGCTAATGCTGACGTGC AAGAACGGCTCCTGCCATATCTGCCTTCCGGCGAATCTTTACCTCAAAGTGGCGAGAAATTCAAAACACTCTTAC TTCACCGCAATTTCAACAAGCTTTAGGAATGTTTCCGCGGCACTCGCTTCTGGACAATTAGGTCCTTTGATGTGTC AATTTGGCCTTCCAGCCGAAGCAGTAGAAGCGGCTAATAAAGGAGACGTAGAGGCCCTTCGCAAAGGCTATGCAAA ATAATGCAAAACCAGAACAAAAGGAAGGGGACACTAAAGATAAGAAAAGATGAGGAAGAAGATATGAGTCTTGAT
scFv(IgG1)	ATGGAAGTACAATTACAACAATCCGGAGCGGAATTGGTGAGGTCTGGCGCGTCCGTGAAGTTAAGCTGTACCGCG AGTGGCTTCAACATAAAGCGCTATGGAATTAATGGGTTAAACAAAAGGCCAGAGCAGGGTCTCGAATGGATCGGC TGGATCAACACTAGGAGTGGCGTGCCGGCGTATGCTCAAGGTTTACCGGAAAGGCGACCATGACTGCAGATACT TCTTCAAACACTGCTTACCTTCAACTTTCATCATTGACCAGTGACGATACCGCGGTATACTACTGCCACGATAAAAC AGAATACTGGGAAGATGGATTTGACGTCTGGGGAGCAGGTACGACCGTAACCGTAAGCTCTGGTGGTGGTGGCTC AGGTGGCGGTGGCTCTGGCGGCGGTGGTTCTGGTGGTGGTTCCGATGTCGTAATGACACAAAACACCCCTCTCTCTC CCTGTATCCCTTGGGGATCAAGCATCAATAAGCTGTTCCGGAAGTAGTTCAAATATTGGGAATTCATACGTTTACTG GTACCTGCAAAAGCCCGGCAATCCCTAAATTGCTGATGTACAGGAACAATAGACGCCCATCTGGGGTGCCAGAC AGATTACGCGGAAGCGGGAGCGGGACAGATTTTACGTTGAAGATTTCAAGGGTCGAGGCAGAAGATCTTGGAGTG TACTTTTTCGCTACCTGGGACGACAGTTTGAGCGGGCGCCTGTTTGGGTCCGGCACCAAGCTCGAGATCAAAAT CATCACCACCATCAGACTACGATATACCCACAACCGCCAGCGAGAATCTGTACTTTCAAGGAGAGTTAGGGATGA GGGGTTCAGCTGGAAGGCGGCTGAAGGTGAAATCCCTGCCCTCTTGTGTTACCGTTTCTAAGATACTGGTAAA AGAAGGTGACACTGTAAAGCTGGTCAAAACAGTTCTGGTGTGAGGCTATGAAAATGGAGACCGAAATTAACGC TCCTACTGACGGAAAAGTTGAAAAGGTGTTAGTTAAGGAAAGAAATGCTGTTCAAGGTGGTCAAGGGTTAATCAA GATCGGCGTTTGA

Data S1. (separate file)

The excel file includes uncropped immunoblot images, and numerical source data underlying the figures.